A Review on EEG-based Multimodal Learning for Emotion Recognition

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Title

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Abstract

Emotion recognition from electroencephalography (EEG) signal is crucial for human-computer interaction, yet poses significant challenges. While various techniques exist for detecting emotions through EEG signals, contemporary studies have explored multimodal approaches as a promising advancement. This paper offers an overview of multimodal techniques in EEG-based emotion identification and discusses recent literature in this area. But these models are computational hungry, which is necessary to address through our research, highlighting the need for further research. A relatively unexplored avenue is combining EEG data with behavioral modalities, considering unpredictable levels of reliability. The suggested review examines the strengths and pitfalls of existing multimodal emotion recognition approaches from 2017 to 2024. Key contributions include a systematic survey on EEG features, exploration of EEG integration with behavioral modalities, and investigation of fusion methods like conventional and deep learning techniques. Finally, key challenges and future research directions in implementing multi-modal emotion identification systems.

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1. Introduction

Human beings undergo emotional reactions in response to various events or situations. The specific emotion a person feels is influenced by the specific circumstance that serves as the trigger. For example, an individual might feel excitement upon achieving a personal goal, while they may experience sadness in the face of a significant loss. Emotions, therefore, reflect our subjective responses to the diverse experiences life presents. Understanding human emotions is of utmost importance when it comes to ensuring effective communication and interaction among individuals as well as between individuals and machines (Dwijayanti et al. 2022). Therefore, the recognition of emotions has been a focal point of extensive research in numerous studies including psychology, neuroscience, and computer science (Zeng et al. 2021). Understanding the role of socioaffective inferential mechanisms in identifying social emotions is essential, given the crucial influence of social context on emotional exchanges (Mumenthaler et al. 2020). Computers have the potential to respond to people in a manner that is emotionally suitable by employing the technique of emotion recognition (Loveys et al. 2022). Furthermore, understanding how socioaffective cues shape emotional recognition is essential for optimizing computer-mediated interactions. Moreover, through emotion recognition, users have the ability to receive feedback regarding their emotional state. This can be highly advantageous for users as it enables them to be more aware of their emotions and the impact they have on their behavior, ultimately aiding in their personal development and self-improvement (Bahreini et al. 2016).

Emotion recognition (ER) is important in various fields, including healthcare, retail and affective computing. It allows for the identification and evaluation of emotional states, which can be useful in medical diagnosis, brain-computer interfaces, and assisting individuals with disabilities. Typically ER often relies on data gathered from facial expressions and auditory cues, and other non-verbal signals such as speech, behavior, and biosignals detected by respiration, and blood volume pulse (Giannakakis et al. 2019). Traditional methods of sensor data collection and labeling often involve exaggerated or intense emotions, which do not reflect real-world scenarios. Therefore, there is a need for more realistic and subtle emotional data for modelling emotional intelligent machines. Studies have shown that multi-modal approaches, combining facial video data, speech, and physiological data can improve the accuracy of ER models (Sebe et al. 2005). These models can help in patient-centered healthcare solutions and contribute to the development of affective computing (Cai et al. 2023). While several papers have reviewed unimodal approaches (Yadav et al. 2022), this paper aims to bridge the gap by offering a comprehensive review of multimodal methodologies, thereby offering a more holistic understanding of the subject matter. In contemporary research, multimodal learning methods, known for their ability to integrate information from various sources, have emerged to capitalize on several benefits: (i) they maximize the correlation between two or more different modalities (ii) they assist machine learning algorithms in understanding and training with multiple forms of input data (ii) they prevent redundant or repetitive information fusion when merging different modalities (iv) they are essential for robust processing, particularly when one modality becomes unreliable.The scope of this draft is shown in Fig.1.

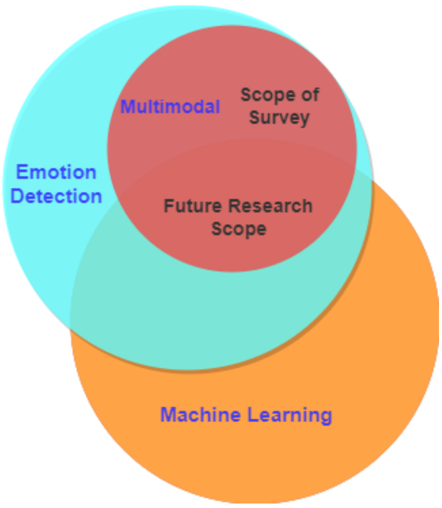

Fig 1 Survey scope and future research direction

Affective computing strives to develop machines (or computers) capable of comprehending and responding to human emotions. In reality, computers may find it challenging to recognize emotions because the internal nature of emotions renders them inaccessible to external machines (van den Broek et al. 2010). A standard process pipeline (Broek 2011) helps overcome this limitation (i) by gathering measurable information about a particular emotional state (indicators) (ii) by adding additional information i.e labels to the gathered data which provide subjective annotations of emotions (annotations), and (iii) by analyzing the relationship between indicators and annotations (modelling). The goal of the modelling, step (iii) is to use the indicators and annotations to predict or determine the user's emotional state. Essentially, the model can learn patterns or correlations in the data to predict true emotional state.

Various methods are employed to gather indicators, such as physiological signals, speech, emotional words, and facial expressions. When it comes to annotations, researchers choose between a discrete emotion-model and continuous emotion-model, depending on whether the annotation scale is discrete or continuous. In the past, emotions were often sorted into distinct categories (Plutchik 1982; Ekman 1992), making it difficult to capture varying levels of emotional intensity (Gatti et al. 2018) and the changing nature of emotional experiences over time.

Moreover, emotions are best understood as continuous phenomena rather than discrete entities, and labels alone cannot accurately convey the intensity or strength of an emotion (Soleymani et al. 2011). However, the use of continuous annotation based on dimensional models, like the circumplex model (Russell 2003; Posner et al. 2005), is now favored. The Russell's Circumplex model is called Valence-Arousal Spatial (VAS) model. The VAS model maps affective concept to a spatial point. Varieties of affective states occupy different points in the two-dimensional space, and distinguished by their spatial distance. For example, emotions like happiness (with high valence and arousal) and sadness (with low valence and arousal) can be distinguished based on their spatial distance within the multi-dimensional space. The valance scale is ranged from negative or unpleasant to pleasant, whereas arousal scale is ranged from active or excited to inactive (Soleymani et al. 2011). The limitations of this approach its inability to discriminate identical emotions. For instance, anger and disgust are both negative-aroused emotions, making them challenging to differentiate using only a two-dimensional approach. Mehrabian (Bakker et al. 2014) expanded the emotion model to 3-dimensions to address this issue. The additional dimension, represented by the dominance axis, reflects an individual's level of control over an emotion.

Current research in the field of emotion identification emphasize dimensional representation (2D) is more congruent with the results obtained from developmental studies of affect. This approach allows for a more precise definition of intensity and takes into account the dynamic and evolving nature of emotional experiences.

Lastly, various frameworks are used for detecting emotions by considering multiple features, and these have been suggested and examined in previous research. However, the accuracy of emotion prediction can be influenced by various factors, such as choosing effective methods for measuring, signal analysis, and extracting features, designing experiments, how participants perceive stimuli, and majorly the algorithms used for modeling. Many authors frequently report these factors, and they have significant implications in numerous studies. How is the process of measuring emotions and preparing input data typically undertaken is challenging and still an ongoing area of research in many studies. Experiments can happen in different settings, and two usual ways are in labs and in real-life situations. It's challenging to thoroughly study the impacts of emotions by purposely making people feel certain ways. People often feel and react based on the emotions of those nearby, showing that our responses are affected by others' feelings. In recent times, as far as the automatic tracking of emotions is concerned, various modeling algorithms have been developed, transitioning from simple shallow methods to more advanced deep neural networks (Liu et al. 2016).

The cutting-edge of research on automatic ER (AER) systems has mainly focused on utilizing combined audio (speech) and visual (faces) signs from video recordings. Besides investigating the distinguishing cues (features), the studies also use several fusion strategies for combining multimodal information to improve emotion detector performance. Beyond audio-visual signs, combined physiological-physical responses shown to be effective for improving multi-modal emotion detector performance (Yang et al. 2023). We describe a multimodal emotion classification (MEC) system pipeline, shown in Fig.2. The pipeline consists of three major components such as the multi-sensor generated data, modality fusion network, and decision making system which classify emotions into various categories. The fundamental procedure is composed of the below four steps:

Step 1: *Identifying Relevant Modalities*
The complementary data sources that can enrich a comprehensive understanding of human emotions are to be determined.

Step 2: *Feature Extraction*
EEG signals provide brainwave patterns, audio yields pitch and tone, video captures facial expressions and facial thermal images, and physiological data reflects bodily responses. This step showcases the extraction of distinct features from each modality.

Step 3: *Modality Fusion*

In this step, the visualization of strategies for fusing modalities is performed to create a model capable of accommodating various types of input data and enabling effective fusion.

Step 4: *Decision Making*
A refined approach to emotion recognition is fostered by utilizing multimodal inputs, including internal and external behavior path, coupled with modelling algorithms such as machine learning techniques.

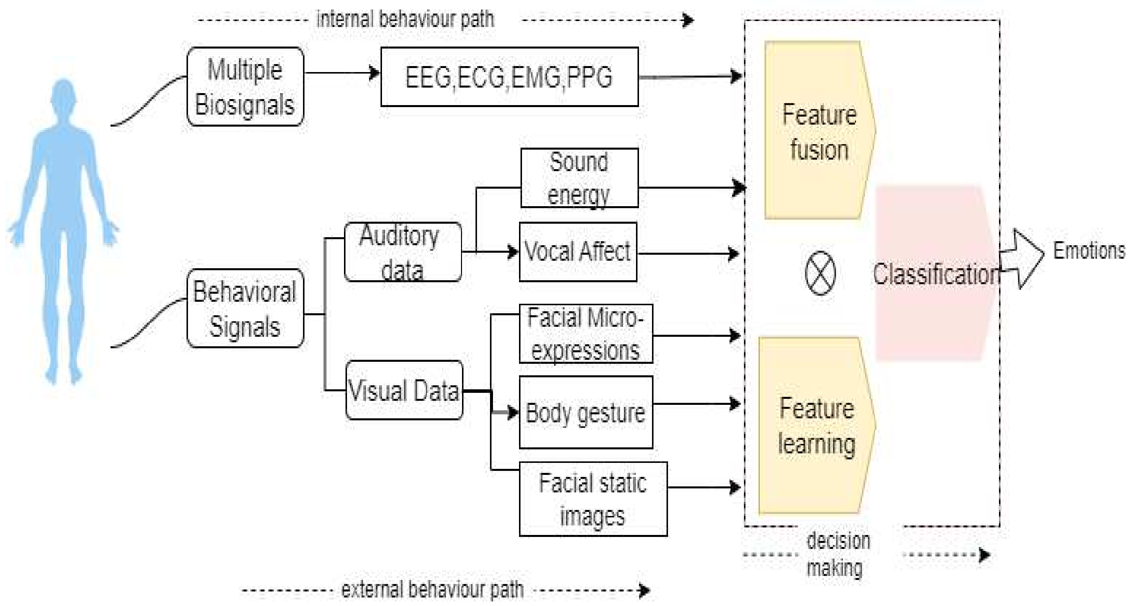

Fig 2 A generic MEC system pipeline

1.1 Biosignal feedback

Multiple data sources possess an incredible ability to convey a broad category of signs related to an individual's feelings; e.g., workload stress, sleepiness, and emotions, which exist within one's mind and are mental states (Eysenck et al. 1994). However, such feelings are not just at psychical level; they also impact our organisms' at physiological level as well (Yakovyna et al. 2021). For instance, strong emotions trigger various bodily reactions; the researchers were carefully tapping into the biosignals to uncover these signs to understand emotions better or to decipher emotions. Researchers today primarily acknowledge two bio-signal measures: one involves the use of peripheral physiological signals (PPS), while the other employs brain electrical activity measures captured from central nervous system (van den Broek et al. 2010) . For emotion recognition, both are equally valuable and complementary methods.

Biosignal feedback (physiological measure) offers a precise means to monitor an individual's emotional state (Kim et al. 2005). There is a growing interest in using biosignals, as opposed to facial or vocal expressions, for detecting affect (Picard 1997). The similarity of physiological responses to emotions across different societies, as well cultures supports the efficacy of this approach (Park et al. 2013). The key advantage lies in their ability to sidestep conscious influences, such as social masking. Modifications in physiological signals linked to emotional states are reflexive and are involuntary, and individuals are normally unaware of these changes (Domínguez-Jiménez et al. 2020). Additionally, in cases where analyzing emotional individuals with verbal disabilities, relying on physical signals like voice may not always be feasible (Bakhshi and Chalup 2021). Moreover, biosignal feedback provides a well-defined emotional state that is essentially undistorted. Currently, there is a trend favoring the adoption of physiological measures, particularly electroencephalography (EEG) based brain wave signals, to quantify inner emotional states. EEG, as a physiological measure, holds advantages in emotion recognition by clearly indicating brain activity. Also, emotion detectors based on EEG has demonstrated good performance (Yang et al. 2018).

Combining brain wave data obtained from EEG signals with details about emotional characteristics in physical (or expressive) responses is a compelling avenue of exploration. The integration of physiological and expressive signals serves the purpose of strengthening the combined information, improving upon the limitations of each modality, and creating a more robust system. However, in the utilization of multimodal signals, a pivotal concern revolves around the methodology of integrating these signals, representing a field of growing interest within the research community. Nevertheless, the integration of EEG data with behavioral modalities (such as visual or auditory cues) has not been extensively explored in a broad spectrum of multimodal studies.

1.2 Related work

Several studies, including those by Yu and Wang (Yu and Wang 2022) and Craik et al. (Craik et al. 2019) have explored methods to achieve high accuracy in emotion recognition. However, these reviews did not adopt a multimodal approaches, leaving room for further exploration and analysis in this area. Samal et al. (Samal and Hashmi 2024) conducted a review on EEG-based emotion recognition, encompassing multimodal analysis. However, the review did not delve into EEG fusion with various modalities. Ahmed et al. (Ahmed et al. 2023) presented a detailed review of multimodal ER using learning algorithms, which also presented available datasets and data acquisition tools, deep learning and conventional classifiers and their application areas.

Baltrušaitis et al. (Baltrušaitis et al. 2018) presented a survey on various fusion techniques and challenges of multimodal learning such as co-learning, mostly discussed modalities are text, image, and audio. However, the review does not provide a detailed comparison of the various multimodal fusion patterns and does not evaluate their performance on benchmark datasets. Nonetheless, the review (Baltrušaitis et al. 2018) prompts the researchers core challenges in multimodal representations using machine learning and can be utilized as a valuable resource by researchers in this area.

Bota et al. (Bota et al. 2019) reviewed the existing studies on AER systems using machine learning techniques to analyze physiological signals. The authors' main focus is to provide discussion on several physiological signals such as ECG, PPG, EEG, GSR etc. that have been used for emotion recognition. Even though, the survey highlights the challenges and limitations of AER using physiological modalities, but does not provide comprehensive analysis of the approaches and techniques used to address these challenges. While the report provides some recommendations for future research, a more detailed discussion of specific methods and techniques for addressing the limitations of physiological signal-based AER would have been beneficial.

He et al. (He et al. 2020) surveyed the use of multiple modalities, such as EEG and other brain imaging signals, in combination with behavioral measures to enhance accuracy of affect-based brain computer interface (aBCI). Additionally, they presented multimodal fusion methods connected to behavior and brain signals, hybrid neurophysiology modalities. On the other hand, is also limited because it does not focus on state-of-the-art models across various modalities. For instance, speech modality is not covered in their review.

Zhang et al. (Zhang et al. 2020) discusses the convergence of machine learning and information fusion for endowing machines with emotion recognition abilities. Various EEG-based methods are reviewed, covering feature extraction, reduction, and ML classifiers. Correlations between EEG rhythms and emotions, as well as brain region-emotion associations, are also explored. The review (Zhang et al. 2020) concludes with a comparison of ML and deep learning algorithms and highlights open issues in this rapidly evolving field. The authors reviewed various fusion techniques in general, whereas we reviewed different fusion techniques and multimodal databases comprehensively. In summary, aim of our survey shares similarities with the related works (Zhang et al. 2020; He et al. 2020), which explore recent progress in fusion techniques and multi-modal contribution with a specific emphasis on the EEG features. The Table 1 compares these related surveys on emotion recognition.

Considering the gaps in existing surveys, this paper offers a thorough examination of the up and downs (pitfalls) associated with current EEG-based multimodal methods. The survey includes all relevant articles published from 2017 to 2024, sourced from reputable databases including PubMed, MDPI, Science Direct, Springer link, IEEE Xplore, and Google Scholar. This survey mainly focuses on fusion of EEG signals, behavioral and other biosignals. Therefore, the main contributions of this survey can be summarized as follows: (i) the presentation of a systematic survey on the fusion of EEG data, (ii) the exploration of use of EEG signals with diverse modalities (speech, text, facial), (iii) the exploration of EEG fusion methods such as conventional and deep learning fusion, and (iv) the provision of insights into open challenges and research guidelines in this domain.

Table 1 Comparison of previous and related surveys and reviews

Ref	**Multi-modal analysis**	**Discussion on EEG fusion with**		**Fusion methods/ operations**	**EEG feature extraction**	**Dataset discussion**	**Multimodal use cases**	**Addressing through RQs**	**Approach to survey/review**
		Behavior signals	**Bio-signals**						
(Yu and Wang 2022)	✗	✗	✗	✗	✓	✗	✗	✗	EEG-based emotion detection
(Craik et al. 2019)	✗	✗	✗	✗	✗	✗	✗	✓	EEG-based emotion detection
(Ahmed et al. 2023)	✓	✗	✗	✓	✗	✓	✗	✓	Emotion acquisition tools; fusion methods
(Samal and Hashmi 2024)	✓	✗	✗	✗	✓	✓	✗	✗	Focus on BCI systems
(Baltrušaitis et al. 2018)	✓	✗	✗	✓	✗	✗	✗	✗	Taxonomy of multi-modal machine learning
(Bota et al. 2019)	✓	✗	✗	✓	✓	✓	✗	✗	Study of emotions; focus on biosignals
(He et al. 2020)	✓	✓	✓	✓	✗	✓	✗	✗	Focus on aBCI system
(Zhang et al. 2020)	✓	✗	✗	✓	✓	✓	✗	✗	Study of emotions; focus on EEG and other biosignals
Ours (-)	✓	✓	✓	✓	✓	✓	✓	✓	Study of emotions; focus on EEG fusion with behavior and other biosignals

2. Methods

To categorize studies and streamline the selection of pertinent papers for this research, a systematic review process called PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) was utilized. The PRISMA process consists of four key sequential steps: paper identification, the elimination of duplicates through scanning, eligibility filtering, and the inclusion of papers to construct a final paper selection list. Fig.3 depicts the steps undertaken.

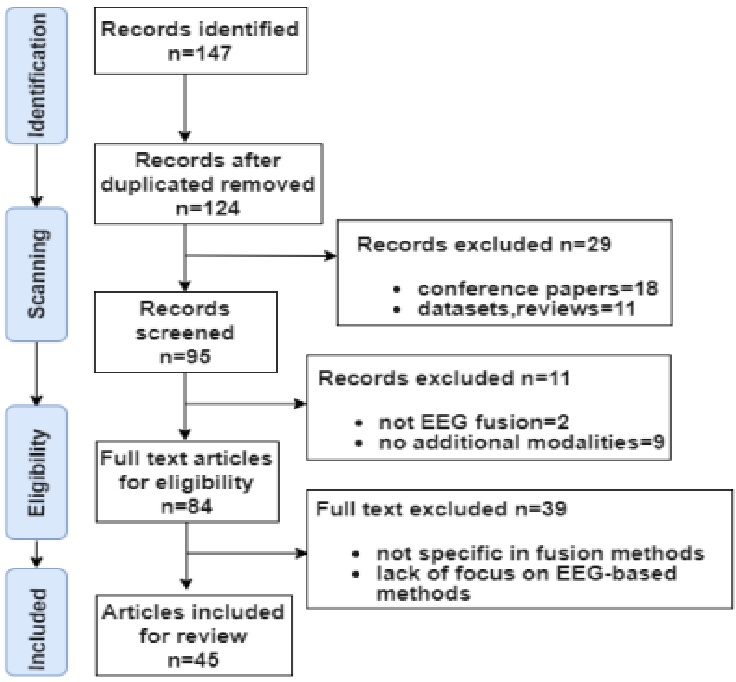

Fig 3 Flow diagram illustrating the paper selection process based on PRISMA (n- represents the number of papers)

A total of 147 articles were retrieved from various databases, including 26 from Science Direct, 72 from IEEE, and 20 from Springer, with additional articles obtained from MDPI, Frontiers, and PubMed via Google Scholar. After excluding 23 duplicated papers and 18 conference papers, further refinement involved eliminating non-related studies, those unrelated to EEG fusion, and articles published before 2017 from the survey. The present study followed steps, as recommended by (Moher et al. 2010)

2.1 Paper Collection

After identifying the research questions, papers from the selected studies were collected. This section covers the search strategy and the criteria used for study selection. To tailor our literature search and focus this review on specific aspects, we formulated the search query using the following terms: a) Emotion Recognition, b) Emotion Detection, c) Electroencephalography/EEG, d) physiological signal fusion, and e) multimodal data fusion.

We conducted a systematic examination of related publications available in major online research repositories, such as IEEE Xplore, Science Direct, SpringerLink, PubMed, and MDPI, to identify the necessary articles. For instance, we conducted a search in the IEEE database with an example query: (("emotion detection") OR ("emotion recognition")) AND ("EEG") AND ("multimodal").

In evaluating the systematic literature review (SLR), we applied inclusion and exclusion criteria to pinpoint relevant primary research works. Our primary focus was on journal papers, with only a minimal consideration of a select few papers from reputed conferences. The Fig. 4 further categorizes the papers according to their respective publication years.

After removing repetitive and irrelevant articles, the selection was refined to 45 articles. We utilized Zotero as a citation management tool to compile the bibliography. The inclusion criteria were strategically incorporated to streamline our search effectively. These criteria consisted of the following: (1) Inclusion of research articles written in English, (2) Inclusion of articles mentioning fusion or feature fusion either in the title or abstract, (3) Requirement for articles to address aspects of data or sensor fusion, and (4) Inclusion of articles within the scope of our domain. As an example, publication titles that referenced multimodal and EEG but did not utilize multiple modalities were excluded from the study (Wu et al. 2022).

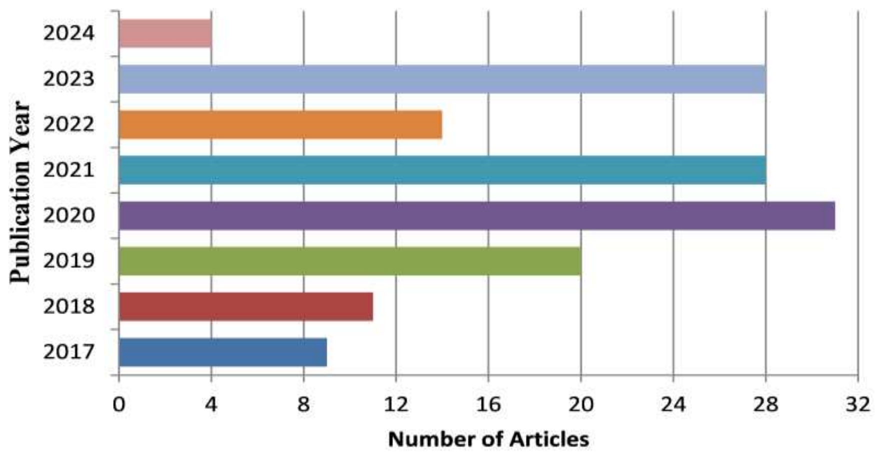

Fig 4 Number of articles per year

3. Research Questions (RQs)

The main survey is structured to address specific research inquiries (RQs), forming the main content of the survey. We have delineated three relevant research inquiries related to the integration of EEG data and behavioral modalities. These research questions are:

RQ1: What is the use of joint EEG and behavioral input to multimodal emotion recognizer? For example, joint measure of Speech and EEG data can effectively detect behavioral dysfunction in medical related studies.

RQ2: What fusion networks are currently accessible, particularly those integrating EEG data?

RQ3: What parameters or features are interesting for accurate emotion recognition?

3.1. RQ1: What is the use of joint EEG and behavioral input to multimodal emotion recognizer?

The utilization of different modalities, coupled with EEG signals, is a significant contributor to the advancement of our understanding of complex cognitive processes and the improvement of efficacy in multimodal ER systems. Through the study of multiple sources of information, a more comprehensive understanding of how the brain processes and responds to stimuli can be gained, ultimately leading to the creation of more accurate and effective emotion recognizer. The review structure of RQ1 comprises three primary components: data modalities, use cases, and datasets. Data modalities are further divided into two sub-components, namely biosignals and combinatory modalities. The investigation into use case components relies on combinatory modalities. The subsequent subsections provide detailed elaboration on the taxonomies and their corresponding components.

3.1.1 Data modalities

The study of data modalities thoroughly examines the relationship or interplay, aiming to discern how diverse forms of data, ranging from physiological signals to individual behavioral expressions, contribute to the broad landscape of human emotions. A thorough investigation on various behavioral modalities including visual modalities (facial and body gestures), audio modality, and text modality was presented by Poria et al., (Poria et al. 2017). This sub-section highlighted various bio-signals.

Peripheral Physiological Signals (PPS)

PPS signals refer to the physiological responses of an individual's body that are linked to psychological processes and emotions. Common psychophysiological measures include: galvanic skin response (GSR), photoplethysmogram (PPG), electrocardiogram (ECG), and facial electromyogram (facial-EMG).

The physiological responses arise from two main components of the nervous system: the Central Nervous System (CNS), encompassing the brain and spinal cord, and the Autonomic Nervous System (ANS), reflecting changes in cardiovascular, electrodermal, and muscular activities. ECG is employed to measure cardiovascular activity, providing parameters such as heart rate (HR) and heart rate variability (HRV). Electrodermal activity (EDA) or GSR gauges skin's electrical conductivity, offering a reliable response to external stimuli. The blood volume pulse is monitored through PPG (Allen 2007), a device in contact with the skin that records changes in cardiac electrical potential over time. Additional physiological indicators of emotional changes include electromyography (EMG), which records muscle electrical activity, as well as respiration rate and body temperature (Bontchev 2016). In a study (Chanel et al. 2007), GSR and EMG demonstrated a high correlation with arousal and valence across all subjects. Irregular respiration and rapid breathing are associated with heightened emotions such as fear.

ECG: Presently, emotion identification systems can be built using a variety of approaches and algorithms. The utilization of ECGs is widespread in many studies, and applications as reviewed in (Hasnul et al. 2021). Emotional experiences can cause changes in the heart rhythm, which can be detected and measured using ECG readings, refer to specific feature from the ECG waveform. Thus making them a useful tool for designing emotion classifier (Brás et al. 2018). In a recent study (Sepúlveda et al. 2021) achieves high accuracy in classifying emotions from wearable device signals, outperforming previous studies with 95.3% accuracy from AMIGOS dataset in a two-dimensional classification.

PPG: Today, PPG signals are gaining popularity in various applications because of their user-friendly and wearable characteristics. PPG signals are derived from the changes in blood volume in peripheral blood vessels, typically measured using sensors placed on the skin surface. The user-friendly aspect implies that obtaining PPG signals does not require invasive procedures, making it effortless and convenient for individuals. Additionally, PPG signals are well-suited for wearable devices, allowing continuous monitoring of physiological parameters, including heart rate and blood oxygen levels, in real-time. Most reported emotion identification methods use PPG signals in a multi-modality approach, but can also be used to detect multiple emotional states using PPG alone (Paul et al. 2023). The proposed algorithm shows superior performance, achieving 97.78% detection accuracy when evaluated on PPG data from the DEAP dataset. However, the existing literature discusses various approaches to create emotional recognition systems; but, there is insufficient exploration of the performance of PPG signals within these systems (Lee et al. 2019). Examining the performance of PPG signals in emotional recognition systems is crucial, as it provides researchers with the opportunity to enhance the precision and efficiency of methods used for characterizing and classifying emotional states based on PPG measures (Goshvarpour and Goshvarpour 2018).

GSR: Another commonly used biofeedback signal for developing automatic emotion classifier is GSR. The signal known as GSR is generated by the electrical activity of the sweat glands and contains valuable information about the sympathetic nervous system. Unlike sensors that generate signals internally, modulating sensors rely on an external source. An example of such a sensor is the GSR sensor, which is a modulating sensor controlled involuntarily and measures skin conductance as mentioned in (Ahmed et al. 2023). GSR measure serves as a remarkably sensitive indicator of emotional arousal (Balconi and Lucchiari 2008). Various studies have employed different GSR-based procedures to identify levels of stress for working professionals (Sriramprakash et al. 2017). In a recent study (Goshvarpour and Goshvarpour 2020) the potential of GSR for emotion recognition is examined, the outcomes demonstrated that the phase space geometry, and consequently, the dynamics of the signal, are influenced by the emotional content of the music video.

EMG: facial-EMG is a biosensor measure utilized to gauge affective empathy by monitoring the activity of facial muscles. This method is based on the premise that individuals tend to mimic the facial expressions of others, providing insights into their emotional experiences (Ferguson and Wimmer 2023). Research using electromyography (EMG) has uncovered that observers tend to mimic expressions of happiness or anger. According to Dimberg et al., 2011, the more empathic individuals exhibit greater muscle contractions in response to these expressions. This finding suggests that highly empathic individuals may possess increased sensitivity to the emotions of others (Dimberg et al. 2011). The identification of emotions using EMG-signals offers the potential for developing a classification system that relies solely on EMG information. The approach presented in (Pereira et al. 2022) provides benefits such as the detection of micro-expressions, consistent signal collection, and eliminates the necessity of acquiring facial images. This research marks an initial stride toward the automatic classification of emotions based exclusively on facial EMG. Additionally, the investigation of emotion sensing using wearable facial-EMG devices to evaluate emotional valence is discussed in [54].

Brain wave (EEG) signals

EEG still remains one of the main utility in clinical neurophysiology and cognitive neuroscience. As per the International Federation of Clinical Neurophysiology (IFCN) Committee, the EEG is characterized as “(1) the science of relating to the electrical activity of the brain”, and “(2) the techniques of recording electroencephalograms” (Steriade et al. 1990). The best aspect that EEG is a real-time display and ongoing indicator of brain activity that offers superior temporal resolution (Thakor and Sherman 2012).

Usually EEG waveforms are recorded using scalp-mounted metal electrodes. During data acquisition electrodes are attached to the subject’s scalp and the potential differences (pds) are recorded. These waveforms are used to interpret the several aspects of the electrical activity arises from human-brain. There are various methods to obtain EEG; one widely used standard method is called 10-20 system (Klem et al. 1999). The standard 10-20 system includes a total of 21 electrodes placed at specific distances of skull, which are determined by measuring the distance between anatomical landmarks (nasion, inion, left preauricular and right preauricular points) on the head. The electrodes are labeled with letters indicating the brain regions they are placed over (Fp:frontoplor, F:frontal, T:temporal, O:occipital, P: parietal, and C:central), and numbers indicating the hemisphere they are located in (“odd numbers for the left hemisphere, even numbers for the right hemisphere”). The parietal and frontal electrode pairs are found to be significant in finding appropriate emotional states (Lin et al. 2010). Ref. (Rojas et al. 2017) outlines the 10-20 system (see Fig. 5). The selection of EEG electrode positions is significant for emotion classification. In Ref. (Zhang et al. 2016b) four electrode (Fp1, Fp2, F3, C4) positions were selected and achieved best performances.

An important issue is selection of appropriate frequency bands: EEG-signal is a blend of waveforms and can be classified based on its magnitude, frequency, spatial distribution, reactivity and wave shape. In particular, frequency band is a common classification method in many studies.. The frequency band of human EEG can be classified into five categories depending on the frequency range and amplitude characteristics: delta (<4 Hz), theta (3-7 Hz), alpha (8-14 Hz), and gamma (>30Hz). Each frequency band is related with specific activities or brain-states. For example, delta (wave frequency is very low) wave observed when a subject is unconscious and in deep sleep, while gamma (very high frequency wave) waves are associated with intense brain activity. Table 2 presents a short description of EEG frequency bands. The EEG frequency bands can provide information about brain-states and are used in several applications including clinical diagnosis to find neurological disorders (Newson and Thiagarajan 2019).

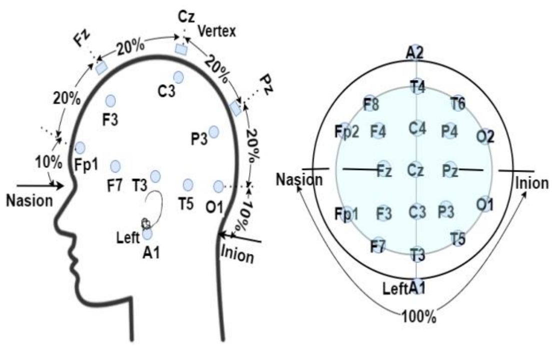

Fig 5 Electrode placement in the EEG system follows the 10-20 international system (Bromfield et al. 2006)

3.1.2 Use cases

EEG-ECG: Physiological measures like ECG and EEG are widely studied for medical diagnosis and biometric recognition. A multimodal biometric recognition system is proposed (Barra et al. 2017), combining ECG and EEG signals, using fiducial (peak) features from ECG and spectrum features from EEG, with good recognition performances. In a recent investigation, researchers are exploring the depth of anesthesia in a clinical setting through the joint measurement of EEG and ECG. This joint measurement approach aims to provide a comprehensive understanding of the patient's physiological responses, particularly focusing on the level of anesthesia (Bahador et al. 2021).

EEG-SPEECH: Researchers are identifying the close relationship between stress and emotions. Emphasizing that stress is often not considered an emotion in traditional emotion models. Especially, long-term stress can lead to cognitive, emotional, and behavioral dysfunctions. The neglect of empathy and stress management in medicine is highlighted. The authors (Duwenbeck and Kirchner 2024) aimed to develop AI-Assistant to detect stress for benefit of various health applications by combing EEG-Speech to enhance both remote and direct healthcare. Nonetheless, a challenge associated with this approach lies in the availability of a comprehensive multi-modal dataset.

EEG-FACIAL: Multimodal emotion recognition, which combines EEG-FACIAL signals, outperforms single modal emotion recognition. This enhanced approach is particularly noteworthy in the context of deaf individuals, where the utilization of both EEG signals and facial expressions in tandem demonstrates superior efficacy compared to relying on a single modality for emotion recognition. The integration of these modalities contributes to a more comprehensive and accurate understanding of emotional states, emphasizing the potential benefits of multimodal approaches in diverse populations (Yang et al. 2022).

EEG-TEXT: Studying customer loyalty involves looking at brain activity, especially in the emerging field of neuromarketing (Kumar et al. 2019) . Researchers use tools like EEG and MEG to understand how our brains respond to products (Pei and Li 2021). Recently, there's a growing trend to combine brain tracking (EEG) with sentiment analysis to detect emotions. An innovative system is proposed (Panda et al. 2020) to combine brain signals and customer reviews (text) to improve how we understand and recognizes emotions during product experiences. The EEG signals and review text have different structures. To make them compatible, both inputs are transformed into a shared feature space. The EEG encoder converts temporal features into a 2D vector, while the text encoder uses a bag-of-words model to extract relevant features, also resulting in a 2D vector. These features from both modalities are then combined into a common feature space to classify emotions. This approach unifies diverse data types to enhance emotion classification.

Table 2 Short description of widely used EEG rhythms and correlation with mental activity ((Kawala-Sterniuk et al. 2021))

EEG rhythms(Hz)	Mental Activity	Location on brain
Delta (0–4)	Sleep, Dreaming-no focus, Unconsciousness	Frontal
Theta(4–8)	Imaginary, Deep relaxation	Midline, Temporal
Low alpha(8–10)	Calm, Wakeful relaxation	Frontal, Occipital
High alpha(10–12)	Self-awareness	
Low beta(12–18)	Active thinking, Problem solving	Frontal, distributed on both sides
High beta(18–30)	Start to alert	
Low gamma(30–50)	Intelligence, self-control	Frontal, Central, Somatosensory cortex
High gamma(50–70)	Cognitive tasks: reading and speaking	

3.1.3 Datasets

To facilitate research in the realm of multimodal emotion learning, numerous datasets have been made publicly available. It's worth noting that a majority of these datasets are typically either multi-physical or multi-physiological in nature. In contemporary times, the scope of multimodal studies has expanded to encompass the combination of various modality elements (EEG and visual components, for example). Table 3 provides details of these datasets, outlining their modality elements and the number of subjects. Few of them are recent and accessible for public use. While most datasets incorporate at least four modalities, some, like DEAP, include as many as nine distinct modalities.

SEED-IV (Zheng et al. 2019) and DEAP (Koelstra et al. 2012) are popular benchmark datasets for studying emotions using physiological signals. SEED-IV collected EEG and Eye movement data from 44 subjects watching video clips inducing sad, fear, happy, and neutral. DEAP consists of peripheral physiological and EEG recordings from 32 subjects viewing one-minute 40-music videos. These datasets provide valuable resources for emotion research in multimodal environments. Both the datasets include recordings of physiological signals, particularly EEG data, which captures brain activity, and peripheral physiological signals. In addition to EEG data, SEED-IV also includes eye movement recordings, while DEAP incorporates multiple peripheral physiological signals alongside the EEG data.

Table 3 Summary of behavioral-physiological databases

Database	**Year**	**Signal type**										**Emotion labels**	**Subjects**
		EEG	ECG	EMG	EOG	GSR	RSP	PPG	Others	Audio	Visual		
DEAP	2012	✓	×	✓	✓	✓	✓	✓	✓	×	✓	arousal, valance, liking	32
MAHNOB - HCI	2012	✓	✓	×	×	✓	×	×	✓	✓	✓	valance, arousal, dominance	27
SEED- IV	2018	✓	×	×	×	×	×	×	✓	×	✓	positive, negative and neutral	44
PME4	2022	✓	×	✓	×	×	×	×	×	✓	✓	anger, disgust, fear, happy, sad, surprise, neutral	11
MED4	2022	✓	×	×	×	×	×	✓	×	✓	✓	happy, sad, angry, neutral	16
DREAMER	2018	✓	✓	×	×	×	×	×	×	×	×	valance, arousal, dominance	23
ASCERTAIN	2018	✓	✓	×	×	✓	×	×	×	×	✓	valance, arousal, dominance, and predictability	58
Phy-MER	2023	✓	×	×	×	×	×	×	✓	×	×	valance, arousal, 7 discrete emotions	30
K-EmoCon	2020	✓	✓	×	×	×	×	✓	✓	✓	✓	valance, arousal	32
AMIGOS	2018	✓	✓	×	×	✓	×	×	✓	✓	✓	valance, arousal, liking	40

The DREAMER dataset (Katsigiannis and Ramzan 2018) encompasses multi-modal data, incorporating EEG and ECG signals captured during affect elicitation via audio-visual stimuli. This database sets a benchmark for affect recognition by leveraging EEG and ECG-based features, yielding results comparable to those obtained from datasets employing costly medical-grade devices. The AMIGOS (Miranda-Correa et al. 2021) dataset includes neuro-physiological signals, video recordings, and annotations of participants' emotions. It will be publicly available for researchers to evaluate the suitability of low-cost devices for affect recognition applications. It allows for the study of affective responses in relation to mood, personality, and social context.

The MAHNOB-HCI (Soleymani et al. 2012) is a widely used dataset for studying affect interfaces in human-computer interaction. It focuses on capturing multimodal data, including physiological signals, audio and visual signals during various interactive scenarios.

ASCERTAIN (Subramanian et al. 2018) is an another multimodal database that connects personality traits and emotional states using physiological responses, containing data from 58 users recorded while viewing affective movie clips. This database contributes to the field of human-computer interaction by addressing the need for agents to recognize and adapt to users' affective states, considering the influence of personality. With the similar motivation a recent study (Pant et al. 2023) presented a Physiological database for Multimodal Emotion Recognition (PhyMER) for studying emotion through physiological response with personality as a context. The PhyMER dataset includes EEG along with other physiological data (EDA, BVP, and skin temperature) collected from 30 subjects. The researchers conducted experiments to examine emotion elicitation in response to video-based stimuli, assessing both inter-rater agreement and the correlation between experienced emotions and personality traits (Pant et al. 2023).

In order to study emotions in social interactions effectively, a novel dataset named K-EmoCon (Park et al. 2020) has been introduced. This multimodal dataset offers detailed annotations of continuous emotions during naturalistic conversations. K-EmoCon comprises diverse measurements, including audiovisual recordings, EEG, and peripheral physiological signals. The data was collected through off-the-shelf devices during 16 sessions, each lasting approximately 10 minutes, featuring paired debates on a social issue. What distinguishes K-EmoCon from previous datasets is its incorporation of emotion annotations from three perspectives: self, debate partner, and external observers.

We identified new datasets MED4 and PME4. The MED4 (Wang et al. 2022) database is a collection of synchronized signals recorded from participants, including EEG, PPG, speech, and facial images. The subjects were exposed to video stimuli specifically created to elicit happiness, sadness, anger, and neutrality. The experiment involved 32 participants and was conducted on different environments: natural noises and an anechoic chamber. The MED4 dataset serves as a resource for researching emotions and their physiological and behavioral correlates in different environmental conditions. Multimodal emotional datasets with physiological data from posed emotions are lacking. While spontaneous expressions offer naturalness, posed ones are less ambiguous and more intense (Chen et al. 2022). A comprehensive dataset named PME4 (Chen et al. 2022) has been created by the researchers. This dataset comprises synchronized physiological and non-physiological signals, facilitating the comparison of emotions expressed by subjects. It encompasses four modalities: EEG, EMG, audio, and video, thus serving as a novel resource for the investigation of posed multimodal emotions.

3.1.4 Challenges

Multimodal ER using EEG faces several challenges: (a) integrating different types of information, (b) finding generalizable features, (c) handling non-stationary signals, and (d) addressing computational requirements.

Firstly, it is difficult to effectively integrate the spatial, spectral, and temporal information of EEG signals to realize better emotion classification performance (Gong et al. 2023). Analyzing non-stationary random signals solely in the time domain isn't sufficient. To address this limitation, fusion feature analysis is introduced, allowing simultaneous consideration of dynamic changes in both time and frequency domains (Zheng et al. 2021). However, this approach comes with added processing steps, making it computationally more demanding than analyzing signals in just one domain (Gao et al. 2020b). This increased computational load could be challenging, particularly in real-time applications and for users without specialized expertise in signal processing techniques.

Secondly, finding discriminative features that can generalize well to different EEG datasets is still a challenge (Taha et al. 2023). Additionally, EEG signals are non-stationary continuous sequential signals, making feature extraction a challenging task (Liu et al. 2023). Furthermore, the increasing amount of computing power required for deep learning poses challenges in providing real-time detection and improving the robustness of deep neural networks (Pan et al. 2023). These challenges necessitate the development of models that can learn spatial and temporal representations of EEG signals, as well as incorporate effective feature fusion mechanisms to improve discrimination power. Researchers have explored various combinations of features and physiological signals, but there is no consensus on which ones are the most informative or effective. This lack of clear evidence suggests that further research is needed to determine the optimal combination of features and physiological signals for ER (Qiu et al. 2018). Moreover, a notable gap exists where model sensitivity and fusion input cues remain underexplored in multimodal studies.

Modalities of varying importance may accommodate differing levels of reliability. This gap prompts a comprehensive literature survey focusing on EEGs' role in multimodal ER.

3.2. RQ2: What fusion networks are currently accessible, particularly those integrating EEG data?

In the context of the multi-modal approach to modality fusion, two key issues stand out as critical: the choice of fusion (fusion level) and the fusion method (how to fuse). The fusion level determines when in the process data from different sources are merged—whether it occurs early on, at the feature level, or later, at the decision level. Different taxonomies are established for fusion architectures; however, they are broadly classified as early, late, and intermediate fusion as illustrated in Fig.6.

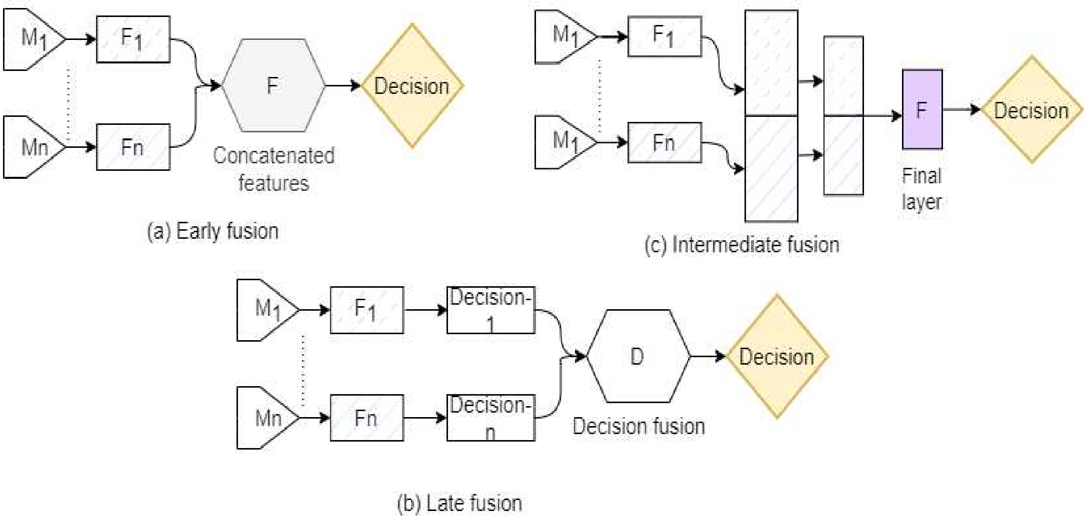

Fig 6 Illustration of different levels of fusion of multi-modal signals

In fusion process, the choice of fusion operation plays a crucial role, determining how data merging occurs, whether through concatenation, weighted fusion, logical operations (max/min/OR), or more advanced techniques like deep learning architectures. These operations exhibit effectiveness in specific contexts and can be categorized as either rule-based or classification-based approaches. Rule-based fusion methods entail basic principles, including weighted fusion, max/min/AND/OR operations, and majority vote rule, while classification-based methods are employed to categorize multimodal observations into predefined classes. Decision fusion, for instance, combines classifier observations linearly with equal weights assigned to each observation class.

However, with the emergence of deep learning, the concept of feature-level fusion has evolved to include intermediate fusion, expanding the possibilities for combining information from multiple modalities (Ramachandram and Taylor 2017). First, we provide basic fusion operations discussed in the literature and next, provide the answers to second research question.

In early-fusion, first low-level descriptors are directly extracted from each modality and are merged using simple concatenation operation before sent to a classification scheme. Usually classification scheme involves a single classifier trained to learn correlations and interactions among low level features of each modality. Another commonly employed merging operation in early-fusion approach is the weighted sum method. This method is frequently utilized in early-fusion scenarios, involving the multiplication of each modality's feature vector by a corresponding weight and then summing them together. We denote the single classifier/model as 'g', and each input signal/modality feature vector as 'X'. Then the final prediction can expressed as:

a. Concatenation

$$X = [X^1, X^2, \dots \dots \dots, X^N] \tag{1}$$

$$Final\ Prediction = g(X)$$

b. Weighted feature fusion

$$X = W^1X^1 + W^2X^2 + \cdots + W^NX^N = \sum_{i=1}^{n} W^iX^i \tag{2}$$

$$Final\ Prediction = g(X)$$

Where, X^i represents individual feature vector extracted. W^i is the weight of the i^{th} feature vector and X represents merged (or final) vector.

The choice of weighting scheme in multimodal fusion significantly affects the fusion process, where weighted sum fusion is a common strategy involving assigning weights to each modality's feature vectors either manually based on prior knowledge or automatically during training to adaptively assign weights based on the data.

Late-fusion often creates multiple independent (called sub-classifiers in this case) models for joint decision making through a range of fusion mechanisms (F) aimed at merging and ensemble the sub-classifier observations into a unified decision. If model g_n is used on modality n (n=1,……,N) the resulting prediction is (Liu et al. 2018):

$$Final\ Prediction = F[(g_1(f_1), g_2(f_2), \dots \dots ., g_n(f_n))] \tag{3}$$

Where $g_n(f_n)$ denotes the outcome of each sub-classifier. $g_n(f_n)$ can be either a confidence score, classification probabilities, classification distance or any other relevant metrics pertaining to various information classes. The decision fusion methods can opt for any of the following mechanisms:

a. Max or Min rule fusion

In the max rule fusion, the final decision is determined by selecting the class with the highest confidence score across all classifiers or modalities. Mathematically it can be represented as:

$$Final\ Prediction = \arg\ max_i \{Conf_i\} \tag{4}$$

In the min rule fusion, the final decision is determined by selecting the class with the lowest confidence score across all classifiers or modalities. Mathematically it can be represented as:

$$Final\ Prediction = \arg\ min_i \{Conf_i\} \tag{5}$$

Where $\{Conf_i\}$ represents the confidence scores (probability or score) associated with $class_i$ from each classifier or modality, and arg max_i (or arg min_i) denotes the index of the class with the maximum (or minimum) confidence score.

b. Majority voting fusion

In the majority voting rule, the final decision is determined by selecting the class that receives the most votes across all classifiers or modalities. Mathematically, it can be represented as:

$$Final\ Prediction = \arg\ max_i \{Votes_i\} \tag{6}$$

Where $\{Votes_i\}$ represents the number of votes received by class$_i$ from each classifier or modality, and arg max denotes the index of the class with maximum number of votes.

c. Decision weighted sum

Another decision-level fusion method, known as "decision weighted sum," entails processing and training each modality with its own classifiers. Afterward, the average prediction scores for each category generate the final outcome. In this method, the fusion process involves weighing the prediction scores for each category from each modality and combining them. The final prediction is determined by the max-rule. To ascertain the best fusion weights, techniques like cross-validation on a validation dataset, reinforcement learning, or beam search methods are usually employed. We denote P_{im} as prediction score of ith modality for the m-th category.

$$Decision\ weighted\ sum = \sum_{i=1}^{n} W_i P_i$$

$$Final\ Prediction = max_m \langle \textstyle\sum_{m=1}^{m} W_{im} P_{im} \rangle \qquad (7)$$

Intermediate fusion involves combining individual modalities at a mid-level processing state before sent to a decision-making system. Intermediate fusion serves as an alternative approach, expanding upon feature-level fusion by integrating deep learning frameworks. This method offers greater flexibility, particularly in facilitating the fusion of representations across various levels of abstraction. Multimodal shared representations can be produced either directly through a dedicated fusion layer or progressively through multiple fusion layers operating at different hierarchical levels (Ramachandram and Taylor 2017). Normally, domain-specific neural networks are employed on various modalities to generate their respective representations, which are then merged or aggregated. A straightforward technique for merging hidden representations involves concatenating weights from diverse modalities. Subsequently, a prediction is typically made based on the aggregated representation, often using another neural network to learn complex mappings between input and output. This characteristic is evident from eqn (2).

The major trends in EEG signal fusion with various affective modalities such as facial, speech, and other PPS signals are the focus of this review section. Various types of fusion approaches have been considered for ER with shallow and deep learning methods. However, the fusion networks reviewed in this article are divided as: conventional (32% studies), deep learning (47% studies), attention-based (9% studies), and autoEncoders-based (12% studies) methods. This categorization is based on their underlying fusion and classification strategies. The conventional approaches implement early and late fusion strategies and uses machine-learning methods. Fig. 7 displays the aggregated information.

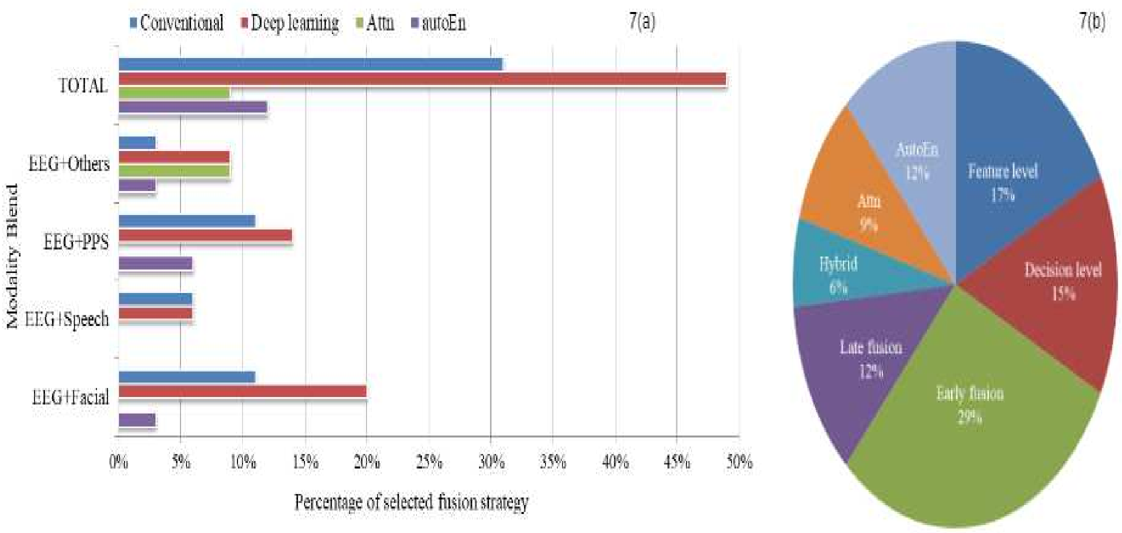

Fig 7 (a) The percentage of selected strategy (b) The agreegated information based on method of fusion stratagies

3.2.1 CONVENTIONAL FUSION METHODS

Conventional fusion methods have been used for combining information from diverse modalities before the emergence of deep learning. They often include: feature-level (early) fusion, decision-level (late) fusion, and hybrid-fusion. In this section, we provide a comprehensive review of these conventional multi-modal fusion strategies for emotion detection reviewed in this paper. Table 4 illustrates the modalities combined and fusion rules employed in the studies reviewed within the conventional fusion category.

Table 4 Compilation of representative works under the conventional fusion category

	Fusion Level	Modality Combination	Fusion Rule	Studies
Conventional Fusion Techniques	Early fusion	EEG+Facial	Concatenation	(Chaparro et al. 2018; Mutawa and Hassouneh 2024),
			Element-wise summation	(Muhammad et al. 2023), (Wang et al. 2023)
			Tensor fusion	(Li et al. 2022)
		EEG+Speech	Concatenation	(Wang et al. 2022), (Li et al. 2021)
		EEG+EyM	Concatenation	(Zhao et al. 2019)
			Element-wise summation	(Fu et al. 2023)
		EEG+Phy	Concatenation	(Zhang et al. 2021b)
		EEG+BVP	Concatenation	(Nakisa et al. 2020)
	Late fusion	EEG+Facial	Weighted sum rule	(Huang et al. 2017), (Lu et al. 2021), (Wang et al. 2021)
		EEG+Speech	Weighted sum rule	(Ghoniem et al. 2019; Wang et al. 2022)
		EEG+EI	Fuzzy fusion rule	(Su et al. 2019)
		EEG+ECG +Speech	Majority voting rule	(Guo et al. 2020)
		EEG+Facial +GSR+PPG	Majority voting rule	(Saffaryazdi et al. 2022)

Note: EyM: Eye movement

Early Fusion: Regarding feature-level fusion, the data acquired from diverse sources are merged or stacked together. Feature-level fusion approaches have been increasingly used to exploit the complementary representation properties among various signals. Several studies have shown that different modalities complement each other in emotion detection for classifying various emotional states.

In Ref. (Zhao et al. 2019) authors adopted a simple feature concatenation strategy between eye movement (representing external subconscious behavior) and EEG (representing internal neural patterns) to ensure the proper representation of information from each modality. The results indicate that five emotions display distinguishable neural patterns, with pupil diameter demonstrating relatively high discriminatory capability compared to other eye movement features.

In Ref. (Fu et al. 2023) the authors proposed a novel early fusion method to effectively combines eye movement signals and EEG signals by extracting complementary information and performing feature fusion using a dual-branch feature extraction module and a multi-scale feature fusion module with cross-channel soft attention.

Researchers have developed various methods for extracting and fusing EEG features, but they often face challenges like high time complexity and insufficient accuracy. In the realm of multimodal ER, achieving high predictive accuracy requires effective techniques for feature extraction and fusion. Deep learning, which emphasizes end-to-end learning, has emerged as a promising approach. Consequently, deep learning models are increasingly employed in multimodal emotion recognition and classification based on EEG data. Due to the dynamic nature and low signal-to-noise ratio of signal data, manually extracted features often fall short in accuracy. To address this, researchers incorporate conventional features alongside manual ones, optimizing their weights based on output accuracy to enhance emotion recognition models. The authors (Zhang et al. 2021b) proposed early fusion method using weighted feature fusion to combine EEG and four different PPS .

The majority of emotion analysis methods, as indicated by references (Chaparro et al. 2018; Li et al. 2022; Wang et al. 2023; Moin et al. 2023; Mutawa and Hassouneh 2024), focus on fusing facial and EEG data to improve the overall recognition accuracy. By combining these two modalities at feature level, the researchers aim to leverage the complementary features provided by brain activity captured through EEG and facial expressions, potentially leading to more robust and accurate ER systems. The work (Li et al. 2022) aims to classify the emotional expressions in individuals with hearing impairment based on EEG signals and facial expressions. In this work, the authors aim to explore the emotional changes in individuals with hearing impairment, utilizing a feature-level fusion strategy that combines EEG topographic maps (ETM) and facial expressions to extract emotional representation features across different modalities effectively. However, the main challenge tackled by existing emotion detection methods in fusing facial expressions is the necessity to train a new model for facial feature extraction, which consumes substantial computing resources. To address this issue a CNN-based feature extraction model combined with weighted fusion method is adopted to enhance multimodal ER by enriching the combined EEG-Facial feature set (Wang et al. 2023).

Poor performance in EEG-based ER models is attributed to the absence of emotion-related information in some EEG data channels. Additionally, it is not recommended to include all EEG data channels in a deep learning model due to concerns about time complexity. Canonical correlation analysis (CCA) is crucial for fusing EEG and facial expressions as it identifies shared patterns across modalities, facilitating the creation of a unified representation for improved emotion recognition accuracy. CCA enables the extraction of correlated features from both EEG and facial data, enhancing the understanding of emotional states by capturing complementary information from multiple sources. The authors (Muhammad et al. 2023) proposed a lightweight deep CCA-based MEC model. By applying DCCA fusion, the method aimed to extract and combine relevant information from the EEG and facial video-clips. The field of video emotion recognition continues to hold significant promise for future exploration (Noroozi et al. 2019). In a study (Xing et al. 2019), the researchers aimed to designing a video emotion classifier by fusing EEG data, and video (both audio, visual) data. Notably, the authors emphasize the importance of video brightness in influencing audience mood and the overall atmosphere within video scenes. To capture this aspect, the luminance coefficient of the video is specifically extracted as a low-level visual feature. The study then proceeds by merging EEG and video features as early fusion, suggesting a comprehensive exploration of both physiological and visual elements in understanding and classifying emotional responses to videos.

The integration of audio signals into EEG responses is an area with limited research. In the study (Wang et al. 2022), the researchers investigated the complementary information between audio and EEG responses using a feature concatenation strategy. This approach allowed them to explore the combined representation of audio and EEG to potentially enhance the efficacy of emotion recognition tasks. Although models based on EEG physiological signals can capture the most authentic human emotional states, the EEG signal's signal-to-noise ratio tends to be low, which can impact emotion recognition accuracy. Given that audio signals also convey emotional information, fusing these two signal modalities may lead to improved emotion recognition effectiveness. The authors of (Li et al. 2021) subscribed to the aforementioned perspective and employed a feature concatenation fusion strategy to fuse EEG and audio-signal representations. The opinion of the researchers in (Jaswal and Dhingra 2023) is that despite feeling and discourse recognition being distinct research domains, integrating voice and EEG signals can lead to a robust psychological model for emotion recognition. They propose exploring novel methods that combine emotion and speech to enhance the accuracy of individual and group emotion detection. They explore the integration of input signals derived from both EEG and audio sources in their analysis as early fusion.

Late Fusion: Regarding decision-level fusion, the outputs obtained from two classifiers in a bi-modal system (combining EEG with facial or speech expressions) are concatenated or fused. EEG is utilized as an inner channel to complement facial and speech expressions, with the goal of improving the overall accuracy and reliability of the ER system. Effective methods for decision-level fusion include averaging /weighted sum of class probabilities (Ghoniem et al. 2019), production rules (Huang et al. 2017) and fuzzy-fusion strategies (Su et al. 2019). When more than two modalities are involved, a multiple decision-making approach based on voting (Guo et al. 2020) is performed to determine the correct emotional category. The category receiving the highest number of votes prevails, and the final decision outcome is determined by the emotion category with the greatest number of votes. This voting-based method helps to integrate the information from multiple modalities and arrive at a more robust emotion classification.

The fusion problem of EEG-Expression combination is addressed using the max-weight fusion strategy, as proposed in reference (Wang et al. 2021). According to the maximum rule, the outcome with the highest score is considered as the final result. In (Lu et al. 2021) a deep-learning based fusion is proposed to combine expression and EEG signal features to solve the problem of bimodal inputs. The main contribution of the research (Saffaryazdi et al. 2022) is centered around the hypothesis that identifying and analyzing the most emotional moments in stimuli allows for a better understanding of the body's reactions to emotions. The primary objective of the fusion in their work is to advance emotion recognition by integrating strategies for analyzing facial micro-expressions with data from EEG, and other physiological signals. A majority voting fusion strategy is adopted for this work.

Hybrid fusion: A hybrid temporal fusion model was used to combine the EEG and BVP signals (Nakisa et al. 2020). The model enabled joint learning and exploration of highly correlated emotional representations across both modalities. The approach involved training each modality with separate deep networks, and then integrating their learned representations to capture the complementary information between BVP and EEG for more effective ER.

In a different study (Cimtay et al. 2020) the researchers conducted experiments using three modalities: EEG, GSR, and facial expressions. For early fusion, EEG and GSR data were combined and input into a deep network. Additionally, a separate classifier was trained for facial expressions. The final emotional state was determined by feeding the outputs of both classifiers into a decision tree.

3.2.2 DEEP LEARNING FUSION METHODS

Within the broad category of multimodal fusion, this section also explores the significant deep fusion techniques including attention and autoEncoders networks. Table 5 illustrates the modalities combined and fusion rules employed in the studies reviewed within the deep fusion category.

Table 5 Compilation of representative works under the deep fusion category

	Fusion Level	**Modality Combination**	**Fusion Rule**	**Studies**
Deeplearning Fusion Techniques	IF	EEG+Facial	IF/LSTM	(Li et al. 2019), (Wu et al. 2020)
		EEG+Phy	IF/LSTM	(Ma et al. 2019)
	AN	EEG+Video	Attention weighted sum	(Choi et al. 2020)
		EEG+EYE+EIG	Attention weighted sum	(Lan et al. 2020), (Zhao et al. 2021)
	AE	EEG+Facial	Shared representations	(Zhang 2020)
		EEG+Eye	Shared representations	(Zhao et al. 2019), (Zheng et al. 2019)

Note: IF: Intermediate Fusion; AN: Attention Network; AE: Auto Encoders

Intermediate fusion: Intermediate fusion is a new paradigm of multi-modal data fusion strategy using deep neural networks, which often known as joint fusion model. The intermediate fusion model primarily involves combining intrinsic feature representations from the hidden layers of deep models with representations from heterogeneous modalities, which are then used as input to a decision function. During fusion process, a shared or joint representation layer (called as fusion layer) is formed where the multi-modal features are merged. The simplest way for merging multi-modal features is to use additive approach (Ma et al. 2019), in which modality-specific semantics are combined in a distinct hidden layer to form a fused vector.

Different from late-fusion method which can model heterogeneous modalities, joint fusion model can properly capture the cross-modality and the inter-modality complementary correlations of multi-modal data. Ref (Ma et al. 2019) presented a joint-fusion method for EEG and PPS signals using a multi-modal “Long Short Term Memory”, LSTM network. The model has a four-layer LSTM network for each modality which enables to capture intra-correlations among EEG and PPG signals, where concatenation of weights or representations occur at each layer during training process. The work of (Li et al. 2019) used a special framework called LSTM to combine information from EEG and facial expressions to classify emotions better. The model allowed understanding how

emotions change over time, making emotion recognition more accurate. In the intermediate fusion approach, recent advancements have integrated typical models such as encoder-decoder architectures to capture shared semantics across EEG and eye movements (Zheng et al. 2019).
Additionally, attention mechanisms have been employed to selectively highlight and combine the most complementary features from EEG and facial video (Wu et al. 2020). This strategy enables the fusion layer to effectively integrate information from multiple sources at an intermediate stage, enhancing the model's ability to learn a cohesive representation of the input data.

Attention-based Fusion: Attention mechanisms have shown promising results across various tasks, such as question answering and machine translation. The attention mechanism enables a neural network to learn adaptable fusion weights for several modalities, leading to enhanced multimodal fusion and identification of emotions (Ahmed et al. 2023). In recent studies, multimodal attention networks have been investigated to enhance the complementary relationship between EEG and other modalities in superior MEC system.

Various modalities may have varying levels of impact on the overall task or prediction of emotions. In an emotion recognition task utilizing facial video modality different types of features may contribute differently to accurately identify an individual's emotional state. Some individuals may experience negative or positive emotions without displaying facial expressions. To accurately identify genuine emotions, incorporating EEG signal alongside facial videos is preferred (Choi et al. 2020). Attention mechanisms allow the model to dynamically weigh the importance of features from each modality, enabling it to selectively integrate the most relevant information from EEG and facial-video modalities to understand person's emotional state. The proposed method features a multimodal attention network, serving as a pivotal module. In this network, the intermediate features extracted from two modality networks are taken, and the weights of each modality are evaluated by comparing their respective features. Subsequently, fusion occurs through the multimodal attention-network during the output stage. Both (Lan et al. 2020) and (Zhao et al. 2021) leverage attention mechanisms to combine information from EEG, raw eye images, and eye movements for multimodal fusion. Ref (Lan et al. 2020) introduced an attention-mechanism among these three modalities and examined the contributions of each modality to emotion recognition. However, ignoring the interaction or mutual influence between modalities may lead to suboptimal fusion results (Zhao et al. 2021). Conversely, in Ref. (Zhao et al. 2021), the attention mechanism is used to improve the accuracy of the model by enhancing the weights of critical feature channels and establishing correlations among different modalities. This is achieved through the novel co-attention layer, which focuses on the most relevant information from each modality and captures inter-modality interactions.

AutoEncoders: Presently, the dual-model fusion strategy employing deep auto-encoders (BADE) has shown promising results in effectively discriminating interactions across EEG and behavioral modalities. In their study, the authors demonstrated the interactions between EEG and facial videos (Zhang 2020) by utilizing automatic encoder structures to extract high-level representations. These representations enabled a deeper understanding of the relationships between different modalities, leading to improved performance in the recognition tasks. However, the extent to which EEG and eye movements offer complementary representations for discriminating between the five emotions (happy, sad, fear, disgust, and neutral) remains unclear and explored in (Zhao et al. 2019). Additionally, eye movement signals not only provide physiological cues but also convey crucial subconscious activities, thereby offering contextual hints for emotion recognition. The authors also employed BADE for modal fusion. During the encoding stage, two Restricted Boltzmann Machines (RBMs) are utilized for processing EEG and eye movement features separately. Subsequently, the outputs of these RBMs, represented by two hidden layers (hEEG and hEye), are concatenated directly as the input for an upper autoencoder. The decoding network mirrors the structure of the encoding network, with the objective of reconstructing the original EEG and eye movement inputs. Lastly, the newly derived shared representations are inputted into a linear Support Vector Machine (SVM) to acquire the recognition results.

In a recent work (Zhou et al. 2018) suggested the use of convolutional Auto-Encoder (CAE) to fuse EEG and multi-type physiological (EP) signals (Zhou et al. 2018) that exhibited a commendable performance. CAEs have the capability to achieve multimodal feature fusion by integrating inputs from various sources, leveraging convolutional neural networks (CNNs).

CAEs differ from conventional autoEncoders (AEs) in that their weights are shared across all locations in the input data, thereby preserving spatial locality (Masci et al. 2011). The trained encoder (seven-layer CNN) of the CAE is utilized to acquire fusion features from multichannel EEG and multi-type EP signals for final result. Finally, Table 6 provides a summary of fusion approaches for multi-modal ER along with databases used, and performance measure in terms of accuracy.

Table 6 Summary of the conventional and deep learning fusion approaches for multi-modal emotion learning.

Article	Data	Fusion level	Database	Classifier	Accuracy (%)
(Wang et al. 2022)	EEG, Speech	early fusion	MED4	ELM	89.65%
(Zhao et al. 2019)	EEG, EyM	early fusion	SEED V	SVM	73.65%
(Chaparro et al. 2018)	EEG, Facial	early fusion	MAHNOB	Random Forest	97.70%
(Ghoniem et al. 2019)	EEG, Speech	late fusion	MAHNOB	FCM-GA-NN	98.53%
(Huang et al. 2017)	EEG, Facial	late fusion	Private data	SVM,FFNN	82.75%
(Su et al. 2019)	EEG, EI	late fusion	Private data	SVM	72.80%
(Guo et al. 2020)	EEG, ECG	late fusion	DEAP,RECOLA	SVC,ELM	89.90%
(Li et al. 2022)	EEG, Facial	early fusion	Private data	CBAM-Resnet34	78.32%
(Li et al. 2021)	EEG, Speech	early fusion	DEAP	Softmax	v:93.2%;a:93.18%
(Muhammad et al. 2023)	EEG, Facial	early fusion	MAHNOB-HCI	Softmax	93.86%
(Zhang et al. 2021b)	EEG, PPS	early fusion	MAHNOB-HCI	HFCNN	89.00%
(Wang et al. 2023)	EEG, Facial	early fusion	DEAP	FCNN	v:96.63%;a: 97.15%
(Mutawa and Hassouneh 2024)	EEG, Facial	early fusion	Private data	LSTM	99.30%

Table 6 continued.

Article	Data	Fusion level	Database	Classifier	Accuracy (%)
(Xing et al. 2019)	EEG, Auditory, Visual	early fusion	Private data	MLP	v:96.79%;a:97.79%
(Jaswal and Dhingra 2023)	EEG, Speech	early fusion	Private data	CNN	94.44%
(Wang et al. 2021)	EEG, Facial	late fusion	Private data	CNN	92.60%
[105	EEG, Facial	late fusion	MAHNOB-HCI	Stacked LSTM	89%
(Li et al. 2019)	EEG, Facial	intermediate fusion	Private data	SVR,LSTM	NA
(Wu et al. 2020)	EEG, Facial	intermediate fusion	DEAP	LSTM	NA
(Saffaryazdi et al. 2022)	EEG, Facial, GSR,PPG	late fusion	Private data	LSTM	v:69.2%;a:70.8%
(Choi et al. 2020)	EEG, Video	attention network	ASIA	FCNN	NA
(Lan et al. 2020)	EEG, Eye movement, EIG	attention network	SEED V	Softmax	82.11%
(Zhao et al. 2021)	EEG, Eye movement, EIG	attention network	SEED	MLP	86.81%
(Zhang 2020)	EEG, Facial	auto encoder	Private data	LIBSVM	85.71%
(Zhou et al. 2018)	EEG,PPS	auto encoder	DEAP	FCNN	92.07%
(Zhao et al. 2019)	EEG, Eye movement	auto encoder	SEED V	SVM	79.70%
(Ma et al. 2019)	EEG, PPS	intermediate fusion	DEAP	MMResLSTM	v:92.30%;a:92.87%

3.3 RQ3: What parameters and methods are interesting for accurate emotion recognition?

Features can be extracted or obtained from time-, frequency-, or time-frequency domains from selected modalities in various ways. Successful approaches for the task of automatic emotion identification include: wavelet transforms, differential entropy, and non-linear dynamics and many more. Fused modalities, fusion level, and the number of parameters dictate the decision strategy for multimodal performance. Fusion level and fused modalities are presented in the previous section (section§3.2). The objective of this section is to understand the number of parameters and methods used in predicting emotions from the selected modalities. In this regard, we delve into the analysis of EEG features, exploring various feature extraction methods and assessing different techniques for feature selection.

3.3.1 EEG Features

The accuracy of AER systems relies on the EEG signals used to describe the emotions. Moreover, better the quality of the descriptor, the more accurate the classification ability. The field of AER uses various EEG features to identify and comprehend different emotional states experienced by individuals. These features can be classified into three categories: time-series features; spectral features; and time-frequency features.

Recently, several non-linear estimators have been utilized to quantify the complexity of EEG signals. The non-linear dynamic features-unlock rich emotion-specific patterns. Below are some of the most commonly used methods for analyzing EEG signals.

a. Time-domain Analysis (TD):

Many researchers attempted TD features to analyze the performance of EEG data to extract the emotion specific patterns. Time-domain features for EEG signals are statistical and descriptive characteristics that are extracted from the raw EEG data in the time domain. These features provide valuable information about the temporal characteristics of brain activity recorded by EEG electrodes.

The popular time-series feature is Hjorth parameters (Hjorth 1970). Hjorth parameters are characteristics derived from the variations in the EEG signal's derivatives (mobility, activity, and complexity). Additionally, the mean absolute value of mobility, activity, and complexity can also serve as features (Wang et al. 2011).

Among the several categories of temporal features, one frequently employed category is referred to as instantaneous statistics (Motamedi-Fakhr et al.). The instantaneous statistical information is directly derived from moments of the EEG signal. The metrics like mean, absolute value, kurtosis, skewness and similar (Wang et al. 2011; Ghoniem et al. 2019) are widely used in many studies. They also encompass measures related to the probability density function of the waveform, such as the mode, median, and entropy. Instantaneous statistics treat each data point within the time series as an independent univariate process, without taking into account any potential correlations or relationships between these individual data points.

Apart from the relatively understandable statistical attributes, we also have the option to derive more sophisticated characteristics from the data. For instance, advanced features like the fractal dimension, FD (such as the Hurst Exponent) is linked to the long-term memory of a time series (Ruiz-Padial and Ibáñez-Molina 2018). Additionally, one can consider the concept of entropy, which measures the level of both regularity and unpredictability in fluctuations across our time series (Wang et al. 2018). Even though statistical metrics less utilized individually, but for emotion identification using physiological data it shows superior performance. For instance the time-domain features of PPG signals exhibit strong correlation with emotional arousal (Fu et al. 2022). Although, time-domain features retain the EEG data's richness, but due to the complex EEG waveform, a universally applicable method for analyzing these features is lacking. Hence, time-domain analysis requires significant expertise and knowledge.

b. Frequency-domain Analysis (FD):

In frequency domain the corresponding affective descriptors are analyzed in terms of its frequency components. For any given time-based signal we have the capability to transform time-series signal into its corresponding spectral representation. This transformation enables us to observe the distinct frequencies that constitute emotion related features.

In the context of spectral analysis, spectral estimation is about determining spectral density (Li et al. 2019). Power spectral density (PSD) or simply spectral density of the signal is the common FD attribute utilized in EEG analysis (Zhang 2019).

The conventional means of EEG signal's spectrum computation involves classic approach, realized through Fourier Transform. On the other hand, PSD computation varies among different studies, majorly categorized as: parametric and nonparametric methods.

Parametric methods assume that the signal can be modeled using a certain equation or model with a set of parameters. For example, parametric methods often involve modeling the signal as an autoregressive (AR) process. Unlike parametric methods, which assume a specific model beforehand, nonparametric methods often rely on the Discrete Fourier Transform (DFT).

The Welch's method stands out among the non-parametric approaches, while the AR method is a prevalent choice among parametric approaches (Zhang 2019). The main drawback of nonparametric methods is that the computation involves data windowing, which can introduce distortion in the resulting power spectral densities (PSDs). In the frequency domain, statistical metrics such as mean, variance, median, standard deviation, skewness, kurtosis, and similar metrics are also frequently employed.

Differential entropy (DE) is concepts from information theory and probability theory, which quantifies the uncertainty and capture the degree of randomness and intricacy present in EEG waveforms. Higher DE values may indicate more intricate or less predictable EEG patterns, which could be linked to certain neurological or cognitive states. In a study (Lan et al. 2020) utilized differential entropy features in EEG based automatic AER system.

c. Time-frequency Domain Analysis (TFD):

When dealing with a non-stationary signal, relying solely on its spectrum becomes less informative. Therefore, it becomes imperative to introduce a time dimension and estimate the signal's frequencies at each time interval (Morales and Bowers). To analyze spectral properties of the non-stationary EEG signals various methods are adopted: power distribution methods and signal decomposition methods. Power distribution methods focus on calculating the distribution of signal power across different frequency bands as the signal evolves over time. Signal decomposition methods involve breaking down a complex signal into its constituent components, enabling the analysis of individual components separately (Zhang 2019).

Short-Time Fourier Transform (STFT) and Wavelet Transform (WT) techniques are used to convert waveform to EEG frequency spectrum. The STFT does not achieve a balanced frequency and time resolution; whereas the WT proves to be better suited for analyzing non-stationary EEG signals (Murugappan et al. 2010).

Wavelet transform is a time-frequency decomposition method. Therefore, wavelet transform emerges as a more appropriate method for breaking down EEG signals into separate frequency bands. Several emotional related features from wavelets (energy and entropy) are generally utilized. However, it might not be as efficient in handling noise components within the signal (Yong and Menon 2015).

Recently empirical mode decomposition (EMD) methods are used in EEG emotion model (Salankar et al. 2021). EMD is a "time-frequency analysis method to deal with non-linear and non-stationary signal" like EEG waves (Zhang et al. 2016c). Hilbert Huang Transform (HHT), Wavelet Packet Decomposition (WPD) (Ting et al. 2008; Zhang et al. 2017b), Matching Pursuit (MP) (Franaszczuk et al. 1998) and Morlet Wavelet Transform (MWT) (Cohen 2019) are also crucial approaches for time-frequency domain analysis. In the field of multimodal ER based on EEG data, commonly utilized feature sets encompass PSD, WT, and DE features. These methods will be detailed in the section §3.3.2.

d. Nonlinear Dynamics:

Recent studies indicate that the human brain behaves as a nonlinear dynamic system, attributing EEG signals to its output (Xingyuan et al. 2009). As a result, researchers are focusing on nonlinear dynamics to scrutinize EEG data (Sharma et al. 2020). Nonlinear dynamics methods can be broadly classified into: chaos theory and information theory. Chaos theory-based approaches involve metrics like correlation dimension (Khalili and Moradi 2009), Kolmogorov entropy (Aftanas et al. 1997), Lyapunov exponent (Übeyli 2010), and Hurst exponent. Information theory methods encompass Approximate Entropy (ApEn) (Pincus 1991), Sample Entropy (SampEn) (Richman and Moorman 2000), Permutation Entropy (PeEn) (Bandt and Pompe 2002), and measures of complexity.

3.3.2 EEG Feature Extraction Methods

The stage after domain analysis is feature extraction. After eliminating noise from the selected modality data, it is crucial to extract important and diverse features before passing them to the classifier. Methods of feature extraction play a critical role in emotion recognition, as they are tasked with capturing information that accurately reflects an individual's emotional state. These extracted features serve as the foundation for algorithms employed in emotion classification tasks. The significance of these methods lies in their ability to determine the accuracy and effectiveness of emotion identification, highlighting the importance of selecting and implementing feature extraction techniques that faithfully represent the nuances of emotional states. The study covers three primary feature extraction techniques, namely WT, STFT, and DE.

a. Methods using STFT

For stationary signals, the Fourier Transform (FT) is often regarded as the most commonly used operation for obtaining the spectrum of the signal. As for non-stationary signals, the same spectrum representation is not purposeful. A spectrogram, on the other hand, is useful to track how different frequencies evolve over time, which is crucial for understanding signal's time-varying phenomena (Brynolfsson 2012). In general, a spectrogram is generated by calculating the periodogram for short-time segments of a signal which is realized through STFT method. As compared to FT the STFT is more localized method which breaks the signal into smaller, overlapping segments using time-window and then applies the Fourier Transform to each of these segments separately (Bruns 2004). A widely used window function in STFT analysis is the Hamming window, as described by Harris in 1978 (Harris 1978).

Welch's method can also be applied to estimate the spectrogram. Welch method uses a modified segmentation scheme and average periodogram. Averaging the periodogram values reduces the statistical oscillations, which generates a stable Welch PSD estimator. Computationally, the periodogram $S_x^l(f)$ is obtained from signal's windowed segment, $X^l(f)$ and window power, U. For *l–th* segment the computations are as follows:

$$X^l(f) = \sum_{n=0}^{N-1} x_l\ (n)w(n)e^{-i2\pi fn} \quad (8)$$

$$U = \frac{1}{N}\sum_{n=0}^{N-1}|w(n)|^2 \quad (9)$$

$$S_x^l(f) = \frac{1}{NU}|X^l(f)|^2 \quad (10)$$

In order to gain PSD with Welch, it is needed the average of modified periodogram, which can be found as:

$$\widetilde{S_x} = \frac{1}{Q}\sum_{l=1}^{Q} S_x^l\,(f) \quad (11)$$

b. Methods using WT

A wavelet is a transient wave function bares some properties such as limited period and finite energy and widely used in digital signal processing. Wavelet transform is an optimal spectral estimation method that replaces any time-series sinusoidal wave as an infinite series of wavelets (known as window function) by translations and dilations. Two distinct WTs are identifiable discrete WT (DWT) and continuous WT (CWT). By transform, a series of elementary functions $\psi_{a,b}$ (t) are generated by shifting and dilating a fixed basis function ψ(t) which is correlated with original signal. The wavelet transform at dilation 'a' and translation 'b' can be described by:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}}\psi\left(\frac{t-a}{a}\right) \quad (12)$$

Where wavelet family and wavelet coefficient are denoted by $\psi(t)$ and $\psi_{a,b}(t)$.

In spectrum estimation using DWT, the signal's decomposition into a set of sub-bands is accomplished through successive high-pass and low-pass filtering of the time-domain signal. The outputs of these filters undergo down-sampling by a factor of 2. This process leads to the generation of the detail component (D) from the high-pass filter, and the other is approximation (A) from the low-pass filter. The DWT decomposition for a given signal $x(t)$ is obtained as (Subasi 2007) :

$$x(t) = \int_{k=-\infty}^{k=+\infty} C_{n,k}\, \varphi(2^{-n}t - k) + \quad \int_{j=1}^{n} \int_{k=-\infty}^{k=+\infty} D_{j,k}\, 2^{-j/2}\psi(2^{-j}t - k) \tag{13}$$

Where $D_{j,k}$ denotes j-th level k-th detail component, $C_{n,k}$ represents k-th approximate component, while $\psi(t)$ and $\varphi(t)$ correspond to wavelet and scaling functions, respectively.

It should be noted that the incorporation of all DWT coefficients from multiple frequency bands into a single feature vector is regarded as impractical and unnecessary (Gandhi et al. 2011). In the reviewed papers, a preference was observed for the utilization of either statistical or non-statistical parameters rather than the direct utilization of wavelet coefficients. For example, the coefficients can be summed to provide estimates of the mean and standard deviation of detailed coefficients. (Subasi 2007). In the reviewed papers, the following are the main DWT features that received significant attention (Table 7).

Table 7 Summary of Time-Frequency (Wavelet-based) Measures

Feature	Formula	Definition
ED(i)	$ED_i = \sum_{k=1}^{N_i} D_{i,j}^2$	Wavelet Energy
R(i)	$R(i) = \frac{ED_i}{\sum_{j=1}^{N} ED_j}$	Wavelet energy ratio
ENT	$ENT = \sum_{i=1}^{N} R(i)[\ln R(i)]$	Wavelet Entropy
μ_i	$\mu_j = \frac{1}{N_j} \sum_{k=1}^{N_j} D_{j,k}$	Mean of detailed coefficients (N_j:j-th level detailed coefficients)
σ_j	$\sigma_j = \sqrt{\frac{1}{N_j} \sum_{k=1}^{N_j} (D_{j,k} - \mu_j)^2}$	Standard deviation of detailed coefficients

c. Methods using DE

In the field of thermodynamics, entropy is a measure of the amount of disorder or randomness in a system. Where as in information theory, which was developed by Claude Shannon, entropy represents the average amount of information or surprise associated with a random variable or a probability distribution (Shannon 1948).

Differential entropy (DE) is used to evaluate the complexity of continuous, and random variables. It quantifies relative or changes in uncertainty, rather than computing an absolute degree of uncertainty (Michalowicz et al. 2013).

Several studies have demonstrated that entropy metrics possess significant potential for studying patterns and regularities within EEG datasets (Acharya et al. 2013). Therefore, the effectiveness of distinguishing various emotions based on their level of irregularity in the EEG signal is demonstrated by entropy (Patel et al. 2021). Distinct kinds of emotions and entropy measures are discussed in the review by Patel et al (Patel et al. 2021). Duan et al. indicated that for a fixed-length EEG sequence, estimating DE is equivalent to the logarithm of the energy spectrum in a specific frequency band (Duan et al. 2013). The fundamental expression of DE can be defined as

$$h(X) = -\int_{-\infty}^{\infty} p(x) log[p(x)]dx \tag{14}$$

Where $p(x)$ represents probability density function (PDF) of a continuous random sample X.

Experimental studies have shown that after band-pass filtering, several sub-frequency bands of EEG-signals closely adhere to the Gaussian distribution $N(\mu,\sigma^2)$, and their associated differential entropy can be determined as

$$h(X) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma_i^2}\right) \log\left[\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma_i^2}\right)\right] dx \quad (15)$$

$$h_l(\mathrm{X}) = \frac{1}{2}\log(2\pi e\sigma_i^2) \quad (16)$$

Where μ is the mean of sample X and σ_i^2 is signal variance. According to Eq. (16), h_l is the DE of the EEG-signal in the i^{th}frequency-band. Here 'e' is the Euler's constant.

However, estimating DE features is quite challenging due to two reasons (i) it requires the estimation of the density of X, and (ii) not suitable for CNN-type classifiers (Hwang et al. 2020) .

Two other common features that can be computed include ratios and difference between DEs -RASM and DASM respectively.

DASM $= h(x_j^{left}) - h(x_j^{right})$ and

RASM $= h(x_j^{left})/h(x_j^{right})$ (17)

These features are used to quantify the asymmetry or differences in the statistical properties of EEG signals between the left and right hemispheres of the brain (Duan et al. 2013).

The idea behind using combinatorial features (a.k.a. mixed domain feature set) is to increase the discriminative power of the feature set which enhance the model's ability to differentiate between different emotional states. Finding the right set of features to achieve the highest accuracy is a challenging task (Wen and Zhang 2017). Combinatorial features can be created by combining features from within the same domain or from different domains. For instance, one can combine frequency-domain features with time-domain features or features within the time-frequency domain can be combined. The choice of which combinatorial features to use often depends on the specific research question, dataset, and the algorithms applied in the analysis.

Certainly, to create combinatorial feature sets, a strategy that combines various feature extraction methods is required (Zhang et al. 2017a). Finally, many literature studies have suggested that by incorporating EEG data into the emotional studies it is possible to identify the inner sense of the emotion of a person. However, the key problem in using EEG data is how to extract the stable and correlated features to form the feature vectors to analyze and find relevant emotions. Feature vectors are "mathematical representation of signals" This implies that during EEG signal analysis, appropriate analytical methods must be employed to ensure meaningful features. The standard quantitative assessment of EEG data is based on linear-analysis, which primarily analyze the signal's characteristics in the time-domain and frequency-domain. The current evolution in the field: some researchers have suggested that "non-linear analysis methods" can provide new insights into how EEG signals can be used for emotion analysis (Sharma et al. 2020).

3.3.2 EEG Feature Reduction/Selection

Feature selection (FS) and Dimensionality reduction (DR) are both methodologies utilized in machine learning and data analysis, yet they serve distinct purposes and employ diverse approaches. FS is a mechanism that selects a best subset of features from the candidate feature set to effectively minimize the feature space based on some criteria, thus enabling machine learning classifiers to operate within the constraints of available computational resources and amplify their predictive accuracy. Unlike other DR methods such as those utilizing projection (like principal component analysis) or compression (such as information theory-based approaches), FS process do not change the "original representation" of the variables. Instead, they simply choose a subset of them, while discarding irrelevant and redundant ones(Saeys et al. 2007).

The FS methods can be grouped into four distinct categories: wrapper methods, filter methods, embedded methods, and hybrid methods (Somol et al. 2010). These categories vary in their methodologies and degrees of interaction with the classification model.

In filter methods, feature selection is dependent on the intrinsic properties and the relevancies that exist among features. Stated differently, filter methods operate independently of any specific learning algorithm (Yu and Liu 2003). In contrast, wrapper methods involve model-specific evaluations and accepting feedback from specific-classifier (Kohavi and John 1997). Similar to wrapper methods, embedded techniques are tailored to specific-classifier. However, embedded models treat FS as an optimization problem, tightly integrating it with a specific-classifier to enhance its performance in classification tasks. The features that are selected are chosen deliberately to have a favourable effect on the current classification task. Hybrid method combines the strengths of more than one approach.

Table 8 presents a general taxonomy of feature selection methods, outlining the key merits and demerits of each technique as specified in Ref. (Saeys et al. 2007).

Table 8 The taxonomy of widely used FS methods and representative works

FS models	Merits	Demerits	Example
Filter	Scaled to high-dimension data, Classifier independence, Computationally simple, Avoids over fitting	Overlooks interaction with classifier	ReliefF(Kononenko 1994) mRMR (Peng et al. 2005)
Wrapper	Simple , Inference with classifier, Relies on model-specific features, Risk of model over fitting	Risk of over fitting, Classifier dependent selection	Recursive feature elimination-RFE (AbdelAal et al. 2018)
Embedded	Inference with classifier, Relies on model-specific features	Classifier dependent selection	Decision tree (Zhang 2020)
Hybrid	Combined strength of Filter - Wrapper schemes	Ignores feature-feature interaction	MIC and QPSO (Chen et al. 2023) RFI-NCA (Tuncer et al. 2021)

Most of the reviewed papers are used linear DR methods like principle component analysis (PCA) in their work. Researchers find nonlinear DR methods particularly useful when they face complex data relationships that linear techniques cannot effectively capture. D.Li et al.(Li et al. 2019) used nonlinear DR technique like t-Distributed Stochastic Neighbor Embedding (t-SNE) to enhance regression performance by removing features that were either irrelevant or redundant.

Apart from DR techniques, FS algorithms ReliefF (Zhang et al. 2016a), mRMR (Atkinson and Campos 2016) have been used for EEG-feature subset selection. Decision tree (DT)-based FS algorithm method used in Ref. (Zhang 2020).

In multi-modal fusion, the redundant features or high dimensionality from processing each modality can hinder the effectiveness of commonly used learning algorithms like neural nets. This necessitates optimization. One way to address this issue is by reducing the input space dimensions to a more manageable size. Clustering emerges as a crucial solution, employed to mitigate this issue by organizing groups of objects or patterns into clusters, thereby reducing the dimensionality of training data as implemented in Ref. (Ghoniem et al. 2019).

Furthermore, we also looked for the FS methods on combined feature set, for the generation of the relevant feature subset. In (Xefteris et al. 2022) the evolutionary approach like genetic algorithm (GA) was used as the combined feature selection method. For comparing number of parameters and extraction methods we selectively chosen the articles based on their works and are listed in Table 9.

Table 9 Summary of parameters selected and techniques used to detect emotions from fused modalities

Ref	EEG features	EEG dimensions (ch x feat)	Fused modality	Feature extraction	Combined Vs Individual FS method
(Ghoniem et al. 2019)	δ,θ,α,β rhythms	14 x 23	Speech	For EEG mixed set of features extracted Speech features are extracted using MFCC, and fundamental frequency	Fuzzy-clustering and GA
(Wang et al. 2022)	δ,θ,α,β, γ rhythms	32 x 5	Speech	160-dimensional differential entropy features obtained for EEG 36-dimensional frame-level MFCC features for speech	ICA
(Su et al. 2019)	δ,θ,α,β, γ rhythms	30 x 6 + 30 x6 + 12 x 5	Eye movement	420-dimensions of EEG features; average energy, and power spectral features 20-dimensional PSD and differential entropy features from eye movement data	Individual / PCA
(Huang et al. 2017)	δ,α1,α2,β1 ,β2,γ1,γ2 rhythms	Not specific	Facial expression	169-dimensional facial image features obtained For EEG, power spectrum density features	Individual / PCA
(Li et al. 2019)	θ,α,β, γ rhythms	14 x 4	Facial expression	Facial geometric features obtained by considering 29 landmarks located at eyes and mouth 56 PSD features from EEG	Individual/ t-SNE

Note: ch: Channels; feat:features

Table 9 Continued

Ref	EEG features	EEG dimensions (ch x feat)	Fused modality	Feature extraction	Combined Vs Individual / FS method
(Zhang 2020)	$\theta,\alpha,\beta,\gamma$ rhythms	64 x 48	Facial expression	14-dimensional WPD features of EEG Sparse representation of facial features	Individual / Decision tree
(Zhao et al. 2019)	$\delta,\theta,\alpha,\beta,\gamma$ rhythms	62 x 5	Eye-movement	310-dimensional DE features of EEG 33 eye movement features, pupil diameter, and event statistics	Combined / AutoEncoder
(Guo et al. 2020)	$\delta,\theta,\alpha,\beta$ rhythms	-	ECG	40-dimensional, 5040 power and central frequency features for EEG is obtained 20-dimensional,1064 parameters from ECG data	Individual/ PCA
(Xing et al. 2019)	$\delta,\theta,\alpha,\beta,\gamma$ rhythms	64 x 5	Auditory and Visual	320-dimensions of PSD features for EEG 88 different acoustic features extracted from video 2-dimensions of color energy, and luminance coefficient as visual features from video	Combined/ PCA

3.3.3 Classifiers and Classification schemes for emotion analysis

Advanced pattern recognition algorithms are crucial in the development of modern classifiers, as they enable the system to identify and categorize complex patterns within vast datasets. These algorithms leverage sophisticated techniques such as neural networks and deep learning to accurately distinguish between different categories, thus enhancing the performance of various tasks in fields of affective computing applications.

In this subsection, we delve into various learning techniques to comprehend different emotion classification schemes. Emotional classifiers can be categorized into two main types: shallow and deep learning (DL) models.

a. Shallow-models

Support Vector Machines (SVM): As a popular machine learning technique SVM classifiers play a crucial role in multimodal emotion classification, offering a powerful tool for distinguishing complex emotional states by leveraging diverse data sources. SVM classifiers are commonly used in multimodal emotion classification because of their ability to handle various types of data and find the optimal discriminant hyperplane that separates different emotion classes (Vapnik 1999). By integrating multiple modalities such as facial expressions, voice intonations, and physiological signals, SVMs can enhance the accuracy of emotion recognition systems (Zhao et al. 2019), (Huang et al. 2017). The classifier works by transforming the input data into a higher-dimensional space where a clear separation between classes can be achieved. This makes SVMs particularly useful in scenarios where emotions are subtle and overlap significantly, ensuring a more precise and reliable classification.

Random Forest: Random Forest is an ensemble learning method that is highly effective for emotion classification tasks. It constructs multiple decision trees during training and outputs (prediction) the mode of the classes of the individual trees (Breiman 2001). Random Forest can handle various types of input data, making it suitable for multimodal emotion classification where it can process facial expressions, voice features, physiological signals, and textual data simultaneously.

One of the strengths of Random Forest is its ability to evaluate the importance of different features. This can be particularly useful in emotion classification to determine which features (e.g., face keypoints) are most indicative of particular emotions (Chaparro et al. 2018). Random Forest can maintain accuracy even when some of the data is missing, which is common in real-world emotion classification scenarios where all sensors or modalities might not always be available.

Fuzzy-c means clustering (FCM): FCM, also known as soft clustering, was developed by Bezdek (1981) using fuzzy set theory (Peizhuang 1983), this method assigns observations to one or more clusters based on similarity measures. The most commonly used similarity measures are distance, intensity, or connectivity. The basic FCM algorithm, being centroid-based, requires a predetermined number of clusters and is highly sensitive to initial centroid placement.

However, many ER problems lack prior knowledge of the optimal number of clusters. To address this, evolutionary approaches such as genetic algorithms (GA) offer alternative optimization methods by employing stochastic principles to evolve clustering solutions (Ghoniem et al. 2019).

Artificial Neural Networks (ANN):

ANNs are computational models inspired by the human brain's neural structure. They consist of interconnected layers of nodes (neurons) that process data in a manner similar to the biological neural networks. ANNs are particularly powerful for pattern recognition tasks due to their ability to learn from large datasets and capture complex, non-linear relationships within the data. The structure of an ANN consists of layers of interconnected neurons: an input layer that receives the data, one or more hidden layers that process the data through weighted connections, and an output layer that delivers the final prediction or classification as illustrated in Fig 8.

In the context of emotion detection, ANNs are utilized to identify and classify emotional states from various data sources such as facial expressions, voice intonations, physiological signals, and textual content. By training on labelled datasets, ANNs can learn to recognize subtle emotional cues and make accurate predictions, making them a valuable tool in the field of affective computing.

Ensemble learning: Ensemble learning is a powerful machine learning technique that combines the predictions from multiple models to improve overall performance and robustness. By aggregating the outputs of various classifiers, ensemble methods can reduce the risk of overfitting, enhance generalization, and increase accuracy.

This approach is particularly important in classification tasks, where diverse models can capture different patterns and nuances in the data, leading to more reliable and precise results.

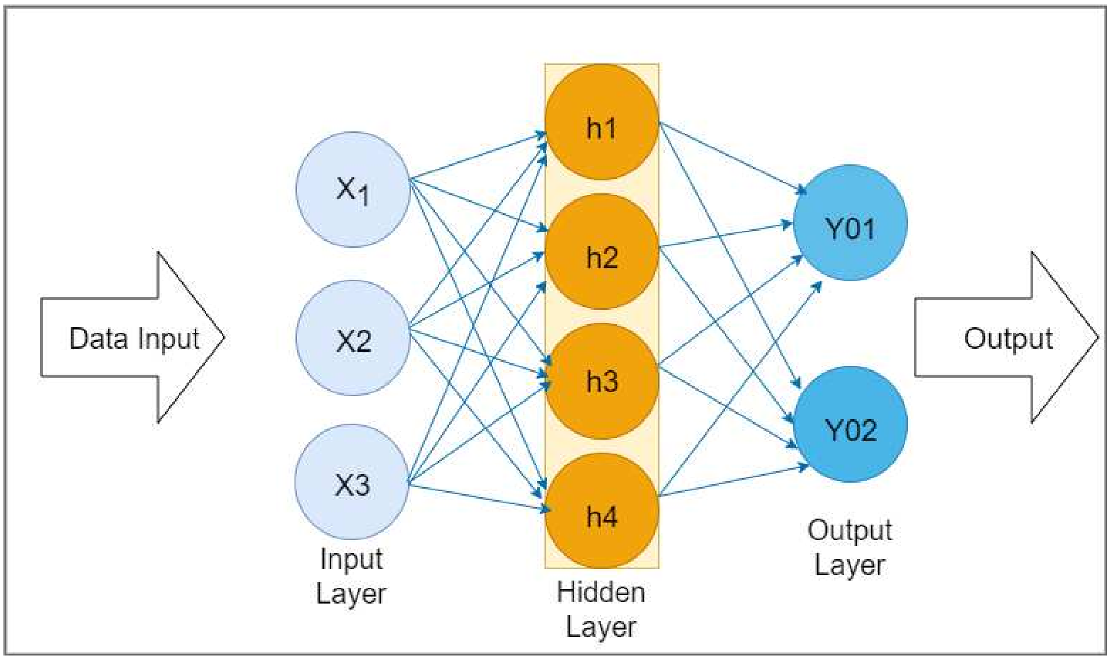

Fig 8 Structural diagram of ANN

Techniques such as bagging, boosting, and stacking are commonly used to create strong ensemble models that outperform individual classifiers.

Today, ensemble learning methods have increasingly garnered the attention of researchers and achieved significant success in ER from combined EEG data. This success is due to their ability to combine the strengths of multiple models, enhancing accuracy and robustness in detecting complex emotional states from brainwave data. Furthermore, ensemble approaches help mitigate the variability and noise inherent in EEG signals, providing more reliable and consistent classification outcomes.

AdaBoost, short for Adaptive Boosting, is an ensemble learning algorithm that aims to improve the accuracy of weak classifiers by combining them into a strong classifier. Its core principle is based on iterative refinement. By iteratively focusing on the hard-to-classify examples, AdaBoost effectively enhances the performance of the overall classifier, making it robust against overfitting and improving its generalization ability. A step-by-step AdaBoost working description is provided below (Freund and Schapire 1997).

-Training samples are selected, and each observation is assigned an initial weight.
-Weak classifiers are trained on the weighted samples.
-Misclassified instances for each weak classifier are identified.
-Update the weights of the correctly and misclassified samples.
-Finally, a strong classifier is obtained by combining the weighted predictions of the weak classifiers.

b. DL-models

DL-models are a subset of ML techniques that use multiple layers (deep architecture), enabling them to learn high-level representations directly from input stream, for instance, sound or image. The practice of using DL-models for emotion identification involves two essential steps: at the core, the model learns modality-specific features from emotional data, which is called "feature or representation learning", and then it applies a classification scheme via appropriate classifier. However, for effective multimodal classification, it is crucial to first extract modality-specific representations before performing fusion. A variety of similar architectures are introduced for this purpose, but their essential components for extracting features specific to each modality may vary (Guo et al. 2019). Here, we will present some of the typical DL-models used for learning monomodal representations.

Deep Belief Net (DBN): DBN is a form of generative graphical deep model. Structurally, the backbone network of DBN uses restricted Boltzmann machine (RBM). The RBMs are a kind of "undirected graphical model", which consists of hidden and visible units. Notably, the RBM model lacks connections between the units within the same layer. A visible layer (or hidden layer) is constructed out of stochastic visible units (or stochastic hidden units).

Typically a DBN is formed by combining a series of given RBMs, referred to as layers. Where, the hidden units of the initial (or bottom) RBM serves as the visible units for the subsequent RBM. As we progress through these layers, the bottom layer captures the basic structure of the actual data, while the subsequent layers extract high-level or more abstract features, as illustrated in Fig 9. However, to more effectively model complex patterns from diverse input modalities, the use of a multimodal DBN is advantageous.

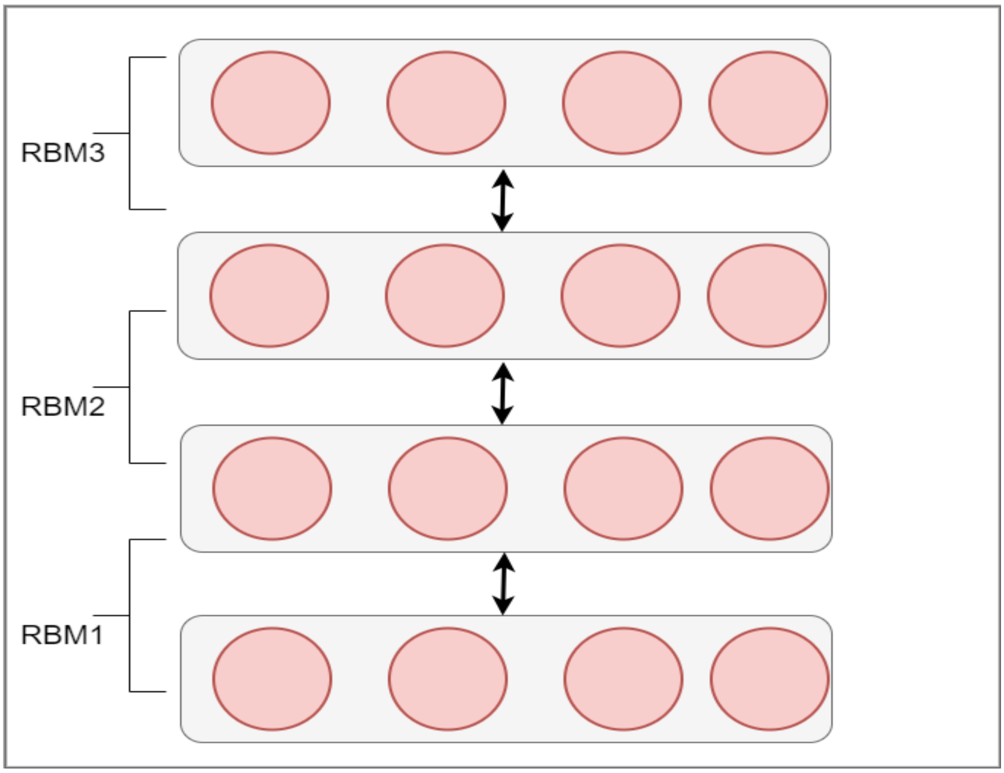

Fig 9 Structural diagram of DBN with three RBMs (Hua et al. 2015)

Convolution Neural Network (CNN): CNNs are also known as ConvNets. ConvNets are exceptionally effective at automatically and efficiently extracting spatial hierarchies of features from visual modalities, which makes them particularly powerful for tasks such as video recognition, and image dehazing.

To extract visual features from images, the essential CNN components are: (a) input layer (b) linearly stacked convolution layers, and (c) fully connected output layer. To analyze EEG signals effectively, a robust CNN model can be employed. First, the EEG signals are converted into visual transformations such as spectrograms. These spectrograms serve as input to the CNN, allowing it to learn hidden representations within the data. By leveraging these hidden representations, the CNN can accurately interpret and classify the underlying patterns in the EEG signals.

Recurrent Neural Networks (RNN): RNN is a kind of DL-model deals with variable-length serial data, which includes sound and time-series data. Mostly, RNNs maps the input activations to the output and then transfers the hidden states to the output through a recurrent feedback connection. Unlike other DL-models, the RNN computing uses both forwad-pass computing and backpropagation-through-time computing. For backpropagation computing, it proceeds to process the current input and hidden states simultaneously. In addition to their effectiveness in monomodal tasks, RNNs have demonstrated their utility in various multimodal problems that necessitate modeling both long- and short-temporal dependencies within input sequences. For example, Wu et al.(Wu et al. 2020) used a stacked LSTM-based network for multimodal ER

Autoencoder (AE): Another key architecture in DL-model is AEs (Rumelhart et al. 1986), which are typically a combination of encoder and decoder modules. Structurally, an encoder-decoder system has three primary components; input, hidden, and output processing layers. The input layer primarily used to obtain raw input. Next, the hidden layer (intermediate units) is utilized to encode raw signal into compressed representations, whereas reconstruction of raw signals occurs at output layer.

These DL-models have recently brought in attention from the multimodal research community because of their significant potential to learn "robust multimodal representations" from diverse modalities, thereby enhancing the performance of multimodal training. For instance, a multimodal encoder-decoder scheme with multiple intermediate layers (called stacked AE) achieves joint compression while preserving correlations among different modalities. In general, autoencoders (AEs) are implemented using various deep learning networks. For instance, the encoder and decoder components can be realized by different neural networks. Based on the concept of encoder-decoder, Masci et al. (Masci et al. 2011) implemented the encoder with CNNs and proposed the Convolutional Autoencoder (CAE), which adopts an unsupervised learning algorithm for encoding. In recent years, CAEs have achieved good results for unsupervised feature extraction. Fig 10 illustrates the structure of the CAE, as mentioned in (Zhou et al. 2018).

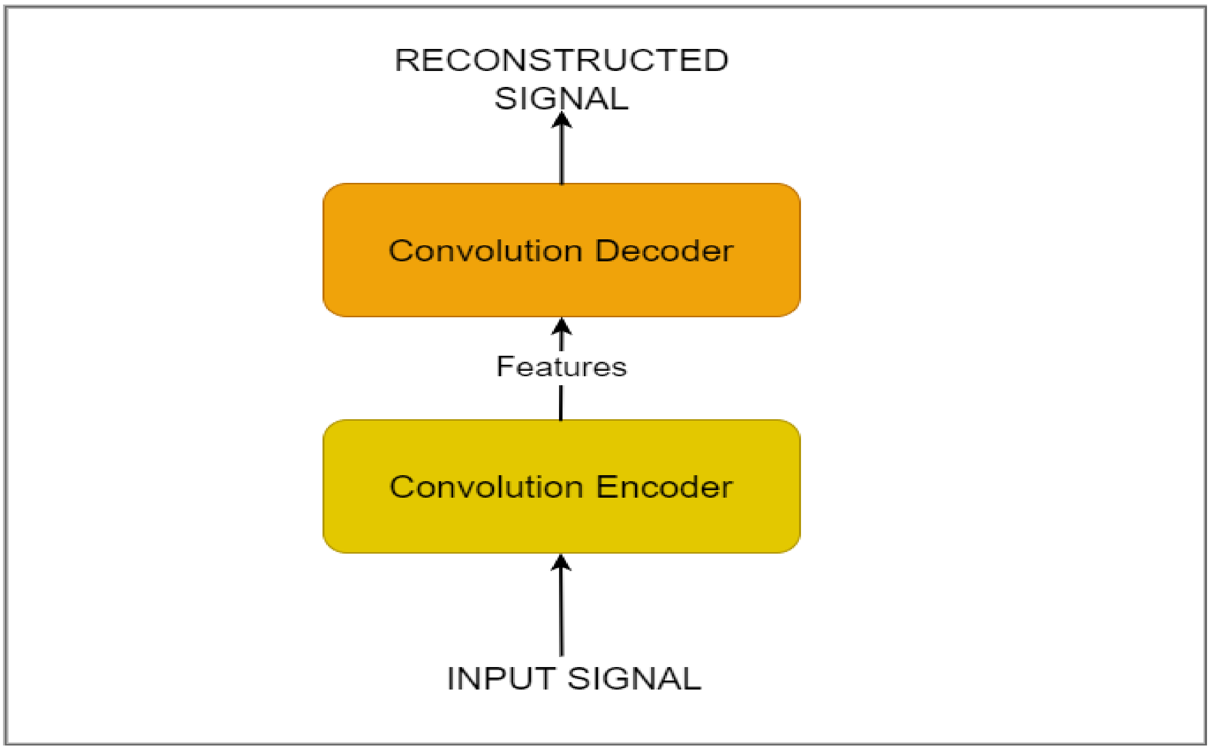

Fig 10 Illustration of Convolution AE (CAE)

c. Multimodal classification schemes

In their study, referenced as (Lan et al. 2020), researchers introduce a novel deep-learning architecture termed "Deep Generalized Canonical Correlation Analysis with an Attention Mechanism" (DGCCA-AM). They emphasize the efficacy of multimodal learning in capitalizing on the complementary aspects of diverse signals, often surpassing the performance of approaches relying on single modalities. The DGCCA-AM model integrates Canonical Correlation Analysis (CCA) to manage multiple signals while incorporating an attention mechanism for adaptive fusion. This fusion mechanism dynamically adjusts weights to exploit generalized correlations across various modalities. Experimental findings reveal the effectiveness of their fusion strategy, achieving an overall classification accuracy and standard deviation of 82.11%, and 2.76% respectively, for the classification of emotions utilizing three modalities.

In (Li et al. 2019), researchers propose a framework that amalgamates EEG and facial data for continuous ER. Employing for late fusion, the framework aims to enhance the understanding of emotional states by leveraging both modalities. In a separate investigation, documented in (Li et al. 2022), the author introduces a novel multimodal fusion strategy, merging EEG signals with facial expressions to classify emotions in individuals with hearing impairment. Experimental findings reveal that after fusion, the mean classification accuracy for emotion recognition reached 78.32%. Notably, this mean accuracy surpassed that achieved using solely facial expressions (67.90%) or EEG signals (69.43%). For the classification task, the study utilized a deepNeuralNet named CBAM_ResNet34.

In (Moin et al. 2023), the study tackles the shortcomings of existing emotion recognition methods by presenting an innovative multi-modal approach that merges EEG and facial data. The method leverages spectral- and statistical- features extracted from EEG signals, amalgamating them with facial image features through three distinct classification schemes: k-nearest neighbor, ensemble methods, and support vector machines. Notably, the proposed approach demonstrates impressive performance, yielding accuracy rates of 97.25% for valence and 96.1% for arousal. These results underscore the efficacy of the multi-modal strategy in enhancing emotion recognition capabilities.

In their work, Zhongjie Li et al. (Li et al. 2021) introduce a unique framework tailored for multi-modal emotion classification. Their approach integrates Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) models to effectively capture both spatial and temporal characteristics inherent in raw EEG data, thereby circumventing the need for manual feature engineering. Empirical results demonstrate the efficacy of the their fusion technique, yielding higher precisions for valence (93.20% ± 2.55%) and arousal (93.18% ± 2.71%), respectively. On another front, Nakisa et al. (Nakisa et al. 2020) delve into the intricate task of emotion recognition utilizing EEG and BVP signals. They propose a temporal multimodal fusion method employing a deep learning algorithm to capture the intricate (non-linear) emotional correlations existing within and between modalities. The ConNet-LSTM model, devised within this framework, achieves a notable accuracy of 71.6% in classifying human emotions into dimensional quadrants. These findings underscore the potential of deep learning methodologies in unraveling the complexities of ER across diverse modalities.

The issue of poor accuracy in traditional emotion-detection methods that rely on facial expressions is addressed in (Wang et al. 2021). The paper proposes a "weighted fusion" for decision-level fusion with CNN. The experimental outcomes exhibit excellent precision, wherein the multi-scale feature extraction network attains an accuracy of 94.4% on the CK+ database.

In their research, Cimtay et al. (Cimtay et al. 2020) introduce a cutting-edge ER system that integrates EEG, facial expressions, and GSR data. The system is designed to address the complexity of human emotions through a hybrid-fusion approach, leveraging Convolutional Neural Networks (CNN). Impressively, this model achieves a notable accuracy of nearly 81.2% on the LUMED-2 database across emotion categories including neutral, happy, and sad. The proposed model is especially valuable for accurately identifying emotional states, even in situations involving natural deceptive facial expressions.

Combining features from several modalities can lead to high- dimensionality and complexity in learning processes for many machine learning algorithms. To address these challenges (Ghoniem et al. 2019), introduced a novel approach using hybrid fuzzy-evolutionary computation methodologies for feature learning and dimensionality reduction. The proposed system focuses on fusing EEG and Speech modalities using multi-hierarchical neural nets. Public databases MAHNOB-HCI and SAVEE were used to test the model and obtained 98.21% and 98.26% accuracies respectively.

4. Open Issues and Discussion

4.1 Open Issues and Further Research

EEG-based multimodal emotion recognition has made significant strides, but several open issues continue to challenge researchers in this field. Addressing these issues is crucial to enrich the accuracy and utility of ER systems in various domains. The following points outline a set of open issues and areas for future investigation within the realm of multi-modal emotion recognition, identified during our comprehensive review.

(1) Multiple Feature Collection: MEC systems commonly initiate by deriving low or high-level attributes subsequent to pretreatment procedures (such as preprocessing blocks). This field of research is still in its developmental stages, with no unanimous agreement on the optimal features to employ for particular tasks. In numerous investigations, practitioners simply aggregate a multitude of attributes to construct a matrix of numbers for classification purposes. If the chosen feature collection fails to accurately capture the inherent emotional cues of the unprocessed signal, the system's performance will suffer, irrespective of the employed algorithm. Consequently, the selection of unique information holds pivotal importance in enhancing overall system efficacy.

(2) Data Fusion Criteria: In the multimodal learning, the task of selecting the optimal fusion scheme is indeed a complex research endeavor. Presently, decisions based on equal weighting or voting mechanisms fail to guarantee the precision of a system. Furthermore, a significant number of studies tend to overlook the distinct and interconnected emotional cues present during early fusion (Zhang et al. 2021a). Due to the nature of a late-level fusion approach, where modalities are integrated after independent processing, the impressive accuracy attained by specific modality might be compromised by the relatively poor accuracy of another modality. Conversely, deep-leaning fusion schemes which are at initial stages, posing ongoing challenges (Gao et al. 2020a). Consequently, the design or selection of an optimal fusion strategy emerges as crucial concerns for the practical systems.

(3) Constrained Dataset with Labels: Scarce set of classified data poses significant challenges in machine leaning (ML) endeavors. The foremost concern is compromised model performance, resulting in suboptimal outcomes. To mitigate the issue, practitioners often resort to co-learning techniques. Co-learning presents a research challenge that involves the necessity of transferring domain knowledge between different modalities utilized with production setting. These challenges become especially significant (Baltrušaitis et al. 2018) in the case of various deep learning (DL) algorithms. Several DL extensions have been proposed to tackle co-leaning issues (Rahate et al. 2022) .

(4) Modality Blend: Much of the research conducted thus far has assessed various hybrid fusion approaches that combine biosensor feedback with external affective measures. However, these studies are also constrained by the scarcity of combinations of selected modalities within real-life contexts. Considerable research efforts may be required to experiment with diverse modality combination (e.g., using the EEG modality with speech and linguistic cues), aiming to achieve more human-like behavior, which stands as an important issue for the future.

(5) Multi-environment Datasets: Having access to annotated multi-modal datasets is essential for the advancement of emotion classification algorithms. While there are numerous multimodal datasets available, there is still a lack of such datasets that incorporate physiological information gathered from large human subjects. Another promising area for future research is the collection of data under multi-environmental (both natural and anechoic) conditions (Wang et al. 2022).

4.2 Discussion

Drawing inspiration from existing literature, it is evident that the analysis of user emotions frequently involves the utilization of audiovisual data within multimodal networks. These networks often employ fusion techniques to capitalize on information derived from both speech signals and temporal cues from video signals. Research demonstrates that integrating audiovisual data significantly improves recognition accuracy. Nevertheless, a considerable portion of multimodal research has yet to explore the amalgamation of EEG physiological data with behavioral modalities such as audio or visual cues. Combining brain wave data obtained from EEG signals with the behavioral manifestations of emotions offers an intriguing avenue for investigation. With this motivation, an extensive literature review is undertaken, focusing specifically on examining the role of EEG data in multimodal emotion identification. Table 6 presents an overview of EEG-based multi-modal models and fusion methodologies employed for ER and classification.

However, a primary challenge in implementing a multi-modal approach for human emotion identification lies in effectively integrating physiological signals with external behavioral cues to accurately identify human emotions. Additional research and innovative approaches are required to overcome this obstacle and advance the field of multimodal emotion recognition.

When the inputs from different modalities are well-aligned in time, a feature-based fusion approach is deemed appropriate. Also feature-level fusion is suitable when the study modalities provide complementary information and have relatively low-dimensionality. It is noteworthy that in cases where the modalities possess varying levels of importance, feature weighting may become necessary as an additional pre-processing step. This technique ensures that each and every modality contributes to the conclusive decision-making process proportionally to its significance in the overall emotion recognition task. By assigning appropriate weights to features from different modalities, the fusion process can be optimized to reflect the relative importance of each source of information, thereby enhancing the accuracy and effectiveness of the multimodal ER system. Decision-level fusion is considered appropriate when independent or redundant information is provided by the modalities, and when different levels of reliability or confidence exist among them, allowing for a weighted combination. In practice, the choice between decision and feature-level fusion is determined by aspects such as the specific utility, the nature of the data, the availability of modalities, and the desired performance goals. Moreover, in certain scenarios, it can be beneficial to utilize a combination of both strategies by employing a hybrid approach. This approach leverages the advantages of both feature-level fusion and decision-level fusion techniques. By integrating features from multiple modalities at an early stage and then combining the outputs of different classifiers or fusion algorithms at a later stage, the hybrid approach aims to capitalize on the strengths of each strategy. This allows for a more comprehensive and robust multimodal ER system that can adapt to diverse data characteristics and task requirements effectively.

Subsequently, the tasks primarily focusing on the selection of merged modalities were categorized into four main groups: facial (34%), speech (11%), peripheral physiological signals (31%), and other modalities (23%). The category labelled as "others" encompassed studies that chose more than two modalities or visual data. Interestingly, a significant proportion of studies (34%) opted to utilize facial data, as illustrated in Fig. 7. Among these, a total of 11% (4 articles) employed conventional and deep learning 20% (7 articles) fusion methods for facial data.

Following facial data, PPS signals were the next prevalent modality (31%), with 11% (4 articles) employing conventional fusion methods and 14% (5 articles) using deep learning fusion. Lastly, another subset of studies integrated speech data. Within this subset, the majority (2 articles) applied deep learning techniques for feature-level fusion. Notably, no articles were found that mentioned the combination of speech signals with attention networks or autoEncoders. Finally, Table 10 summarizes key studies on multimodal emotion analysis databases.

Table 10 Comparative table: key studies on multimodal emotion analysis datasets

Datasets Used	**Ref.no**	**Emotional states**	**Accuracy (%)**
MED4	(Wang et al. 2022)	happy, sad, angry, neutral	89.65%
CASIA	(Guo et al. 2020)	angry, happy, sad, fear	95.16%
SEED IV	(Zhao et al. 2019)	happy, sad, fear, disgust, and neutral.	79.71%
DEAP	(Zhang et al. 2021b)	valence, arousal	valence 84.7% arousal 83.2%
MAHNOB-HCI	(Xing et al. 2019)	anger, disgust, fear	89%
PME4	(Chen et al. 2022)	happiness, sadness, surprise, neutral	-
PhyMER	(Pant et al. 2023)	disgust, happy, angry, neutral, sad, surprise, fear	NA
K-EmoCon	(Park et al. 2020)	valence, arousal	-

Note:'-' refers no research found for multimodal studies; NA: Not applicable

Given the promising potential of integrating EEG data with behavioral modalities like audio and visual cues, future research should delve into more comprehensive EEG-based multimodal ER systems.

- Future research may explore fusion techniques for disparate modalities in multimodal-based emotion identification systems.
- Have a more diversity in the selection of modalities beyond facial and speech such as text and electrophysiological data.
- More comprehensive information about feature selection methods may be included.
- Explore benchmark datasets that encompass a unique combination of modalities, encouraging and fostering comparisons between different approaches.

- Methods for designing ML and DL classifiers may be further included in subsequent investigations.

5. Conclusions

In this review, we have examined the potential of utilizing two distinct modalities, EEG and behavioral data, for emotion recognition. We have summarized various strategies aimed at enhancing the accuracy of emotion recognition while also addressing potential pitfalls associated with these approaches. Looking ahead, the integration of these viable methods and the utilization of machine learning for emotion assessment hold immense promise, particularly across diverse domains such as advertising, autonomous systems, and smart healthcare applications. Our paper has shed light on the process of uncovering emotions by leveraging features from EEG and behavioral modalities seems to be extremely useful combination in medical-IOT and education systems. We have also discussed and compared two feature fusion methods, providing insights into their effectiveness for emotion recognition tasks. Overall, this review underscores the importance of multimodal approaches in emotion identification and emphasizes the need for sustained exploration and innovation in this field to unlock its full potential for practical utility.

Author contributions R.P wrote the main manuscript text and prepared all figures and tables; U.S checked the details of the full manuscript. All authors reviewed the manuscript.

Data Availability Data availability does not apply to this research.

Declarations

All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

References

AbdelAal MA, Alsawy AA, Hefny HA (2018) EEG-Based Emotion Recognition Using a Wrapper-Based Feature Selection Method. In: Hassanien AE, Shaalan K, Gaber T, Tolba MF (eds) Proceedings of the International Conference on Advanced Intelligent Systems and Informatics 2017. Springer International Publishing, Cham, pp 247–256

Acharya UR, Vinitha Sree S, Swapna G, et al (2013) Automated EEG analysis of epilepsy: A review. Knowledge-Based Systems 45:147–165. https://doi.org/10.1016/j.knosys.2013.02.014

Aftanas LI, Lotova NV, Koshkarov VI, et al (1997) Non-linear analysis of emotion EEG: calculation of Kolmogorov entropy and the principal Lyapunov exponent. Neuroscience Letters 226:13–16. https://doi.org/10.1016/S0304-3940(97)00232-2

Ahmed N, Al Aghbari Z, Girija S (2023) A systematic survey on multimodal emotion recognition using learning algorithms. Intelligent Systems with Applications 17:200171

Allen J (2007) Photoplethysmography and its application in clinical physiological measurement. Physiol Meas 28:R1. https://doi.org/10.1088/0967-3334/28/3/R01

Atkinson J, Campos D (2016) Improving BCI-based emotion recognition by combining EEG feature selection and kernel classifiers. Expert Systems with Applications 47:35–41. https://doi.org/10.1016/j.eswa.2015.10.049

Bahador N, Jokelainen J, Mustola S, Kortelainen J (2021) Multimodal spatio-temporal-spectral fusion for deep learning applications in physiological time series processing: A case study in monitoring the depth of anesthesia. Information Fusion 73:125–143. https://doi.org/10.1016/j.inffus.2021.03.001

Bahreini K, Nadolski R, Westera W (2016) Towards multimodal emotion recognition in e-learning environments. Interactive Learning Environments 24:590–605. https://doi.org/10.1080/10494820.2014.908927

Bakhshi A, Chalup S (2021) Multimodal Emotion Recognition Based on Speech and Physiological Signals Using Deep Neural Networks. In: Del Bimbo A, Cucchiara R, Sclaroff S, et al. (eds) Pattern Recognition. ICPR International Workshops and Challenges. Springer International Publishing, Cham, pp 289–300

Bakker I, van der Voordt T, Vink P, de Boon J (2014) Pleasure, Arousal, Dominance: Mehrabian and Russell revisited. Curr Psychol 33:405–421. https://doi.org/10.1007/s12144-014-9219-4

Balconi M, Lucchiari C (2008) Consciousness and arousal effects on emotional face processing as revealed by brain oscillations. A gamma band analysis. International Journal of Psychophysiology 67:41–46. https://doi.org/10.1016/j.ijpsycho.2007.10.002
Baltrušaitis T, Ahuja C, Morency L-P (2018) Multimodal machine learning: A survey and taxonomy. IEEE transactions on pattern analysis and machine intelligence 41:423–443
Bandt C, Pompe B (2002) Permutation entropy: a natural complexity measure for time series. Phys Rev Lett 88:174102. https://doi.org/10.1103/PhysRevLett.88.174102
Barra S, Casanova A, Fraschini M, Nappi M (2017) Fusion of physiological measures for multimodal biometric systems. Multimed Tools Appl 76:4835–4847. https://doi.org/10.1007/s11042-016-3796-1
Bontchev B (2016) Adaptation in Affective Video Games: A Literature Review. Cybernetics and Information Technologies 16:3–34
Bota PJ, Wang C, Fred AL, Da Silva HP (2019) A review, current challenges, and future possibilities on emotion recognition using machine learning and physiological signals. IEEE Access 7:140990–141020
Brás S, Ferreira JHT, Soares SC, Pinho AJ (2018) Biometric and Emotion Identification: An ECG Compression Based Method. Frontiers in Psychology 9:
Breiman L (2001) Random Forests. Machine Learning 45:5–32. https://doi.org/10.1023/A:1010933404324
Broek E van den (2011) Affective Signal Processing (ASP): Unraveling the mystery of emotions. https://doi.org/10.3990/1.9789036532433
Bromfield EB, Cavazos JE, Sirven JI (2006) Slide 14, [10/20 System of EEG Electrode Placement]. https://www.ncbi.nlm.nih.gov/books/NBK2510/figure/A44/. Accessed 3 Jun 2024
Bruns A (2004) Fourier-, Hilbert- and wavelet-based signal analysis: are they really different approaches? Journal of Neuroscience Methods 137:321–332. https://doi.org/10.1016/j.jneumeth.2004.03.002
Brynolfsson J (2012) Time Frequency Analysis of EEG Measured When Performing the Flanker Task
Cai Y, Li X, Li J (2023) Emotion Recognition Using Different Sensors, Emotion Models, Methods and Datasets: A Comprehensive Review. Sensors 23:2455. https://doi.org/10.3390/s23052455
Chanel G, Ansari-Asl K, Pun T (2007) Valence-arousal evaluation using physiological signals in an emotion recall paradigm. In: 2007 IEEE International Conference on Systems, Man and Cybernetics. pp 2662–2667
Chaparro V, Gomez A, Salgado A, et al (2018) Emotion Recognition from EEG and Facial Expressions: a Multimodal Approach. In: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). pp 530–533
Chen J, Ro T, Zhu Z (2022) Emotion Recognition With Audio, Video, EEG, and EMG: A Dataset and Baseline Approaches. IEEE Access 10:13229–13242. https://doi.org/10.1109/ACCESS.2022.3146729
Chen W, Cai Y, Li A, et al (2023) EEG feature selection method based on maximum information coefficient and quantum particle swarm. Sci Rep 13:14515. https://doi.org/10.1038/s41598-023-41682-5
Choi DY, Kim D-H, Song BC (2020) Multimodal attention network for continuous-time emotion recognition using video and EEG signals. IEEE Access 8:203814–203826
Cimtay Y, Ekmekcioglu E, Caglar-Ozhan S (2020) Cross-Subject Multimodal Emotion Recognition Based on Hybrid Fusion. IEEE Access 8:168865–168878. https://doi.org/10.1109/ACCESS.2020.3023871
Cohen MX (2019) A better way to define and describe Morlet wavelets for time-frequency analysis. NeuroImage 199:81–86. https://doi.org/10.1016/j.neuroimage.2019.05.048
Craik A, He Y, Contreras-Vidal JL (2019) Deep learning for electroencephalogram (EEG) classification tasks: a review. J Neural Eng 16:031001. https://doi.org/10.1088/1741-2552/ab0ab5
Dimberg U, Andréasson P, Thunberg M (2011) Emotional Empathy and Facial Reactions to Facial Expressions. Journal of Psychophysiology 25:26–31. https://doi.org/10.1027/0269-8803/a000029
Domínguez-Jiménez JA, Campo-Landines KC, Martínez-Santos JC, et al (2020) A machine learning model for emotion recognition from physiological signals. Biomedical Signal Processing and Control 55:101646. https://doi.org/10.1016/j.bspc.2019.101646
Duan R-N, Zhu J-Y, Lu B-L (2013) Differential entropy feature for EEG-based emotion classification. In: 2013 6th International IEEE/EMBS Conference on Neural Engineering (NER). IEEE, pp 81–84
Duwenbeck R, Kirchner EA (2024) Auditive Emotion Recognition for Empathic AI-Assistants. Künstl Intell 1–6. https://doi.org/10.1007/s13218-023-00828-3
Dwijayanti S, Iqbal M, Suprapto BY (2022) Real-Time Implementation of Face Recognition and Emotion Recognition in a Humanoid Robot Using a Convolutional Neural Network. IEEE Access 10:89876–89886
Ekman P (1992) An argument for basic emotions. Cognition & emotion 6:169–200
Eysenck MW, Ellis AW, Hunt EB, Johnson-Laird PNE (1994) The Blackwell dictionary of cognitive psychology. Basil Blackwell
Ferguson HJ, Wimmer L (2023) A psychological exploration of empathy. In: Conversations on Empathy. Routledge, pp 60–77

Franaszczuk PJ, Bergey GK, Durka PJ, Eisenberg HM (1998) Time–frequency analysis using the matching pursuit algorithm applied to seizures originating from the mesial temporal lobe. Electroencephalography and Clinical Neurophysiology 106:513–521. https://doi.org/10.1016/S0013-4694(98)00024-8
Freund Y, Schapire RE (1997) A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting. Journal of Computer and System Sciences 55:119–139. https://doi.org/10.1006/jcss.1997.1504
Fu B, Gu C, Fu M, et al (2023) A novel feature fusion network for multimodal emotion recognition from EEG and eye movement signals. Front Neurosci 17:. https://doi.org/10.3389/fnins.2023.1234162
Fu Z, Zhang B, He X, et al (2022) Emotion recognition based on multi-modal physiological signals and transfer learning. Front Neurosci 16:1000716. https://doi.org/10.3389/fnins.2022.1000716
Gandhi T, Panigrahi B, Anand S (2011) A comparative study of wavelet families for EEG signal classification. Neurocomputing 74:3051–3057. https://doi.org/10.1016/j.neucom.2011.04.029
Gao J, Li P, Chen Z, Zhang J (2020a) A Survey on Deep Learning for Multimodal Data Fusion. Neural Computation 32:829–864. https://doi.org/10.1162/neco_a_01273
Gao Q, Wang C, Wang Z, et al (2020b) EEG based emotion recognition using fusion feature extraction method. Multimed Tools Appl 79:27057–27074. https://doi.org/10.1007/s11042-020-09354-y
Gatti E, Calzolari E, Maggioni E, Obrist M (2018) Emotional ratings and skin conductance response to visual, auditory and haptic stimuli. Sci Data 5:180120. https://doi.org/10.1038/sdata.2018.120
Ghoniem RM, Algarni AD, Shaalan K (2019) Multi-Modal Emotion Aware System Based on Fusion of Speech and Brain Information. Information 10:239. https://doi.org/10.3390/info10070239
Giannakakis G, Grigoriadis D, Giannakaki K, et al (2019) Review on psychological stress detection using biosignals. IEEE Transactions on Affective Computing 13:440–460
Gong L, Li M, Zhang T, Chen W (2023) EEG emotion recognition using attention-based convolutional transformer neural network. Biomedical Signal Processing and Control 84:104835. https://doi.org/10.1016/j.bspc.2023.104835
Goshvarpour A, Goshvarpour A (2018) Poincaré's section analysis for PPG-based automatic emotion recognition. Chaos, Solitons & Fractals 114:400–407. https://doi.org/10.1016/j.chaos.2018.07.035
Goshvarpour A, Goshvarpour A (2020) The potential of photoplethysmogram and galvanic skin response in emotion recognition using nonlinear features. Phys Eng Sci Med 43:119–134. https://doi.org/10.1007/s13246-019-00825-7
Guo H, Jiang N, Shao D (2020) Research on Multi-modal Emotion Recognition Based on Speech, EEG and ECG Signals. In: Qian J, Liu H, Cao J, Zhou D (eds) Robotics and Rehabilitation Intelligence. Springer, Singapore, pp 272–288
Guo W, Wang J, Wang S (2019) Deep multimodal representation learning: A survey. Ieee Access 7:63373–63394
Harris FJ (1978) On the use of windows for harmonic analysis with the discrete Fourier transform. Proceedings of the IEEE 66:51–83. https://doi.org/10.1109/PROC.1978.10837
Hasnul MA, Aziz NAA, Alelyani S, et al (2021) Electrocardiogram-Based Emotion Recognition Systems and Their Applications in Healthcare—A Review. Sensors 21:5015. https://doi.org/10.3390/s21155015
He Z, Li Z, Yang F, et al (2020) Advances in Multimodal Emotion Recognition Based on Brain–Computer Interfaces. Brain Sciences 10:687. https://doi.org/10.3390/brainsci10100687
Hjorth B (1970) EEG analysis based on time domain properties. Electroencephalography and clinical neurophysiology 29:306–310
Hua Y, Guo J, Zhao H (2015) Deep Belief Networks and deep learning. In: Proceedings of 2015 International Conference on Intelligent Computing and Internet of Things. pp 1–4
Huang Y, Yang J, Liao P, Pan J (2017) Fusion of Facial Expressions and EEG for Multimodal Emotion Recognition. Computational Intelligence and Neuroscience 2017:e2107451. https://doi.org/10.1155/2017/2107451
Hwang S, Hong K, Son G, Byun H (2020) Learning CNN features from DE features for EEG-based emotion recognition. Pattern Anal Applic 23:1323–1335. https://doi.org/10.1007/s10044-019-00860-w
Jaswal RA, Dhingra S (2023) Empirical analysis of multiple modalities for emotion recognition using convolutional neural network. Measurement: Sensors 26:100716. https://doi.org/10.1016/j.measen.2023.100716
Katsigiannis S, Ramzan N (2018) DREAMER: A Database for Emotion Recognition Through EEG and ECG Signals From Wireless Low-cost Off-the-Shelf Devices. IEEE Journal of Biomedical and Health Informatics 22:98–107. https://doi.org/10.1109/JBHI.2017.2688239
Kawala-Sterniuk A, Browarska N, Al-Bakri A, et al (2021) Summary of over Fifty Years with Brain-Computer Interfaces—A Review. Brain Sciences 11:43. https://doi.org/10.3390/brainsci11010043

Khalili Z, Moradi MH (2009) Emotion recognition system using brain and peripheral signals: using correlation dimension to improve the results of EEG. In: 2009 International Joint Conference on Neural Networks. IEEE, pp 1571–1575

Kim J, André E, Rehm M, et al (2005) Integrating information from speech and physiological signals to achieve emotional sensitivity. In: Ninth european conference on speech communication and technology

Klem GH, Lüders HO, Jasper HH, Elger C (1999) The ten-twenty electrode system of the international federation. The international federation of clinical neurophysiology. Electroencephalogr Clin Neurophysiol Suppl 52:3–6

Koelstra S, Muhl C, Soleymani M, et al (2012) DEAP: A Database for Emotion Analysis ;Using Physiological Signals. IEEE Transactions on Affective Computing 3:18–31. https://doi.org/10.1109/T-AFFC.2011.15

Kohavi R, John GH (1997) Wrappers for feature subset selection. Artificial Intelligence 97:273–324. https://doi.org/10.1016/S0004-3702(97)00043-X

Kononenko I (1994) Estimating attributes: Analysis and extensions of RELIEF. In: Bergadano F, De Raedt L (eds) Machine Learning: ECML-94. Springer, Berlin, Heidelberg, pp 171–182

Kumar S, Yadava M, Roy PP (2019) Fusion of EEG response and sentiment analysis of products review to predict customer satisfaction. Information Fusion 52:41–52. https://doi.org/10.1016/j.inffus.2018.11.001

Lan Y-T, Liu W, Lu B-L (2020) Multimodal emotion recognition using deep generalized canonical correlation analysis with an attention mechanism. In: 2020 International Joint Conference on Neural Networks (IJCNN). IEEE, pp 1–6

Lee MS, Lee YK, Pae DS, et al (2019) Fast Emotion Recognition Based on Single Pulse PPG Signal with Convolutional Neural Network. Applied Sciences 9:3355. https://doi.org/10.3390/app9163355

Li D, Liu J, Yang Y, et al (2022) Emotion Recognition of Subjects with Hearing Impairment Based on Fusion of Facial Expression and EEG Topographic Map. IEEE Transactions on Neural Systems and Rehabilitation Engineering 31:437–445

Li D, Wang Z, Wang C, et al (2019) The Fusion of Electroencephalography and Facial Expression for Continuous Emotion Recognition. IEEE Access 7:155724–155736. https://doi.org/10.1109/ACCESS.2019.2949707

Li Z, Zhang G, Dang J, et al (2021) Multi-Modal Emotion Recognition Based On deep Learning Of EEG And Audio Signals. In: 2021 International Joint Conference on Neural Networks (IJCNN). pp 1–6

Lin Y-P, Wang C-H, Jung T-P, et al (2010) EEG-based emotion recognition in music listening. IEEE Transactions on Biomedical Engineering 57:1798–1806

Liu K, Li Y, Xu N, Natarajan P (2018) Learn to Combine Modalities in Multimodal Deep Learning. https://doi.org/10.48550/arXiv.1805.11730

Liu W, Zheng W-L, Lu B-L (2016) Emotion Recognition Using Multimodal Deep Learning. In: Hirose A, Ozawa S, Doya K, et al. (eds) Neural Information Processing. Springer International Publishing, Cham, pp 521–529

Liu Z-T, Hu S-J, She J, et al (2023) Electroencephalogram Emotion Recognition Using Combined Features in Variational Mode Decomposition Domain. IEEE Transactions on Cognitive and Developmental Systems 15:1595–1604. https://doi.org/10.1109/TCDS.2022.3233858

Loveys K, Sagar M, Billinghurst M, et al (2022) Exploring Empathy with Digital Humans. In: 2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW). pp 233–237

Lu Y, Zhang H, Shi L, et al (2021) Expression-EEG bimodal fusion emotion recognition method based on deep learning. Computational and Mathematical Methods in Medicine 2021:1–10

Ma J, Tang H, Zheng W-L, Lu B-L (2019) Emotion Recognition using Multimodal Residual LSTM Network. In: Proceedings of the 27th ACM International Conference on Multimedia. Association for Computing Machinery, New York, NY, USA, pp 176–183

Masci J, Meier U, Cireşan D, Schmidhuber J (2011) Stacked Convolutional Auto-Encoders for Hierarchical Feature Extraction. In: Honkela T, Duch W, Girolami M, Kaski S (eds) Artificial Neural Networks and Machine Learning – ICANN 2011. Springer, Berlin, Heidelberg, pp 52–59

Michalowicz JV, Nichols JM, Bucholtz F (2013) Handbook of Differential Entropy. CRC Press

Miranda-Correa JA, Abadi MK, Sebe N, Patras I (2021) AMIGOS: A Dataset for Affect, Personality and Mood Research on Individuals and Groups. IEEE Transactions on Affective Computing 12:479–493. https://doi.org/10.1109/TAFFC.2018.2884461

Moher D, Liberati A, Tetzlaff J, Altman DG (2010) Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. International Journal of Surgery 8:336–341. https://doi.org/10.1016/j.ijsu.2010.02.007

Moin A, Aadil F, Ali Z, Kang D (2023) Emotion recognition framework using multiple modalities for an effective human–computer interaction. J Supercomput 79:9320–6349. https://doi.org/10.1007/s11227-022-05026-w

Morales S, Bowers ME Time-frequency analysis methods and their application in developmental EEG data - ScienceDirect. 54:101067
Motamedi-Fakhr S, Moshrefi-Torbati M, Hill M, et al Signal processing techniques applied to human sleep EEG signals—A review - ScienceDirect. Biomedical Signal Processing and Control 10:21–33
Muhammad F, Hussain M, Aboalsamh H (2023) A Bimodal Emotion Recognition Approach through the Fusion of Electroencephalography and Facial Sequences. Diagnostics 13:977. https://doi.org/10.3390/diagnostics13050977
Mumenthaler C, Sander D, Manstead ASR (2020) Emotion Recognition in Simulated Social Interactions. IEEE Transactions on Affective Computing 11:308–312. https://doi.org/10.1109/TAFFC.2018.2799593
Murugappan M, Ramachandran N, Sazali Y (2010) Classification of human emotion from EEG using discrete wavelet transform. Journal of biomedical science and engineering 3:390
Mutawa AM, Hassouneh A (2024) Multimodal Real-Time patient emotion recognition system using facial expressions and brain EEG signals based on Machine learning and Log-Sync methods. Biomedical Signal Processing and Control 91:105942. https://doi.org/10.1016/j.bspc.2023.105942
Nakisa B, Rastgoo MN, Rakotonirainy A, et al (2020) Automatic Emotion Recognition Using Temporal Multimodal Deep Learning. IEEE Access 8:225463–225474. https://doi.org/10.1109/ACCESS.2020.3027026
Newson JJ, Thiagarajan TC (2019) EEG Frequency Bands in Psychiatric Disorders: A Review of Resting State Studies. Frontiers in Human Neuroscience 12:521
Noroozi F, Marjanovic M, Njegus A, et al (2019) Audio-Visual Emotion Recognition in Video Clips. IEEE Transactions on Affective Computing 10:60–75. https://doi.org/10.1109/TAFFC.2017.2713783
Pan J, Fang W, Zhang Z, et al (2023) Multimodal Emotion Recognition based on Facial Expressions, Speech, and EEG. IEEE Open J Eng Med Biol 1–8. https://doi.org/10.1109/OJEMB.2023.3240280
Panda D, Chakladar DD, Dasgupta T (2020) Multimodal System for Emotion Recognition Using EEG and Customer Review. In: Mandal JK, Mukhopadhyay S (eds) Proceedings of the Global AI Congress 2019. Springer, Singapore, pp 399–410
Pant S, Yang H-J, Lim E, et al (2023) PhyMER: Physiological Dataset for Multimodal Emotion Recognition With Personality as a Context. IEEE Access 11:107638–107656. https://doi.org/10.1109/ACCESS.2023.3320053
Park B-J, Jang E-H, Chung M-A, Kim S-H (2013) Design of Prototype-Based Emotion Recognizer Using Physiological Signals. ETRI Journal 35:869–879. https://doi.org/10.4218/etrij.13.0112.0751
Park CY, Cha N, Kang S, et al (2020) K-EmoCon, a multimodal sensor dataset for continuous emotion recognition in naturalistic conversations. Sci Data 7:293. https://doi.org/10.1038/s41597-020-00630-y
Patel P, R R, Annavarapu RN (2021) EEG-based human emotion recognition using entropy as a feature extraction measure. Brain Inf 8:20. https://doi.org/10.1186/s40708-021-00141-5
Paul A, Chakraborty A, Sadhukhan D, et al (2023) A simplified PPG based approach for automated recognition of five distinct emotional states. Multimed Tools Appl 83:30697--30718. https://doi.org/10.1007/s11042-023-16744-5
Pei G, Li T (2021) A Literature Review of EEG-Based Affective Computing in Marketing. Frontiers in Psychology 12:602843
Peizhuang W (1983) Pattern Recognition with Fuzzy Objective Function Algorithms (James C. Bezdek). SIAM Rev 25:442. https://doi.org/10.1137/1025116
Peng H, Long F, Ding C (2005) Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. IEEE Transactions on Pattern Analysis and Machine Intelligence 27:1226–1238. https://doi.org/10.1109/TPAMI.2005.159
Pereira L, Brás S, Sebastião R (2022) Characterization of Emotions Through Facial Electromyogram Signals. In: Pinho AJ, Georgieva P, Teixeira LF, Sánchez JA (eds) Pattern Recognition and Image Analysis. Springer International Publishing, Cham, pp 230–241
Picard RW (1997) Affective computing. MIT Press, Cambridge, Mass
Pincus SM (1991) Approximate entropy as a measure of system complexity. Proceedings of the National Academy of Sciences 88:2297–2301. https://doi.org/10.1073/pnas.88.6.2297
Plutchik R (1982) A psychoevolutionary theory of emotions. Social Science Information 21:529–553. https://doi.org/10.1177/053901882021004003
Poria S, Cambria E, Bajpai R, Hussain A (2017) A review of affective computing: From unimodal analysis to multimodal fusion. Information Fusion 37:98–125. https://doi.org/10.1016/j.inffus.2017.02.003
Posner J, Russell JA, Peterson BS (2005) The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. Development and psychopathology 17:715–734

Qiu J-L, Liu W, Lu B-L (2018) Multi-view Emotion Recognition Using Deep Canonical Correlation Analysis. In: Cheng L, Leung ACS, Ozawa S (eds) Neural Information Processing. Springer International Publishing, Cham, pp 221–231

Rahate A, Walambe R, Ramanna S, Kotecha K (2022) Multimodal Co-learning: Challenges, applications with datasets, recent advances and future directions. Information Fusion 81:203–239. https://doi.org/10.1016/j.inffus.2021.12.003

Ramachandram D, Taylor GW (2017) Deep Multimodal Learning: A Survey on Recent Advances and Trends. IEEE Signal Processing Magazine 34:96–108. https://doi.org/10.1109/MSP.2017.2738401

Richman JS, Moorman JR (2000) Physiological time-series analysis using approximate entropy and sample entropy. American Journal of Physiology-Heart and Circulatory Physiology 278:H2039–H2049. https://doi.org/10.1152/ajpheart.2000.278.6.H2039

Rojas GM, Alvarez C, Montoya C, et al (2017) Multimodal Study of Resting-State Functional Connectivity Networks using EEG electrodes position as seed. bioRxiv 167585

Ruiz-Padial E, Ibáñez-Molina AJ (2018) Fractal dimension of EEG signals and heart dynamics in discrete emotional states. Biological Psychology 137:42–48. https://doi.org/10.1016/j.biopsycho.2018.06.008

Rumelhart DE, Hinton GE, Williams RJ (1986) Learning representations by back-propagating errors. Nature 323:533–536. https://doi.org/10.1038/323533a0

Russell JA (2003) Core affect and the psychological construction of emotion. Psychological Review 110:145–172. https://doi.org/10.1037/0033-295X.110.1.145

Saeys Y, Inza I, Larrañaga P (2007) A review of feature selection techniques in bioinformatics. Bioinformatics 23:2507–2517. https://doi.org/10.1093/bioinformatics/btm344

Saffaryazdi N, Wasim ST, Dileep K, et al (2022) Using Facial Micro-Expressions in Combination With EEG and Physiological Signals for Emotion Recognition. Frontiers in Psychology 13:864047

Salankar N, Mishra P, Garg L (2021) Emotion recognition from EEG signals using empirical mode decomposition and second-order difference plot. Biomedical Signal Processing and Control 65:102389. https://doi.org/10.1016/j.bspc.2020.102389

Samal P, Hashmi MF (2024) Role of machine learning and deep learning techniques in EEG-based BCI emotion recognition system: a review. Artif Intell Rev 57:50. https://doi.org/10.1007/s10462-023-10690-2

Sebe N, Cohen I, Huang TS (2005) Multimodal emotion recognition. In: Handbook of Pattern Recognition and Computer Vision. WORLD SCIENTIFIC, pp 387–409

Sepúlveda A, Castillo F, Palma C, Rodriguez-Fernandez M (2021) Emotion Recognition from ECG Signals Using Wavelet Scattering and Machine Learning. Applied Sciences 11:4945. https://doi.org/10.3390/app11114945

Shannon CE (1948) A mathematical theory of communication. The Bell System Technical Journal 27:379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Sharma R, Pachori RB, Sircar P (2020) Automated emotion recognition based on higher order statistics and deep learning algorithm. Biomedical Signal Processing and Control 58:101867. https://doi.org/10.1016/j.bspc.2020.101867

Soleymani M, Koelstra S, Patras I, Pun T (2011) Continuous emotion detection in response to music videos. In: 2011 IEEE International Conference on Automatic Face & Gesture Recognition (FG). pp 803–808

Soleymani M, Lichtenauer J, Pun T, Pantic M (2012) A Multimodal Database for Affect Recognition and Implicit Tagging. IEEE Transactions on Affective Computing 3:42–55. https://doi.org/10.1109/T-AFFC.2011.25

Somol P, Novovicová J, Pudil P (2010) Efficient feature subset selection and subset size optimization. In: Pattern Recognition Recent Advances. IntechOpen Rijeka, Croatia

Sriramprakash S, Prasanna VD, Murthy OVR (2017) Stress Detection in Working People. Procedia Computer Science 115:359–366. https://doi.org/10.1016/j.procs.2017.09.090

Steriade M, Gloor P, Llinas RR, et al (1990) Basic mechanisms of cerebral rhythmic activities. Electroencephalography and clinical neurophysiology 76:481–508

Su Y, Li W, Bi N, Lv Z (2019) Adolescents Environmental Emotion Perception by Integrating EEG and Eye Movements. Frontiers in Neurorobotics 13:46

Subasi A (2007) EEG signal classification using wavelet feature extraction and a mixture of expert model. Expert Systems with Applications 32:1084–1093. https://doi.org/10.1016/j.eswa.2006.02.005

Subramanian R, Wache J, Abadi MK, et al (2018) ASCERTAIN: Emotion and Personality Recognition Using Commercial Sensors. IEEE Transactions on Affective Computing 9:147–160. https://doi.org/10.1109/TAFFC.2016.2625250

Taha B, Hwang DY, Hatzinakos D (2023) EEG Emotion Recognition Via Ensemble Learning Representations. In: ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). pp 1–5

Thakor NV, Sherman DL (2012) EEG signal processing: Theory and applications. In: Neural Engineering. Springer, pp 259–303
Ting W, Guo-zheng Y, Bang-hua Y, Hong S (2008) EEG feature extraction based on wavelet packet decomposition for brain computer interface. Measurement 41:618–625. https://doi.org/10.1016/j.measurement.2007.07.007
Tuncer T, Dogan S, Subasi A (2021) EEG-based driving fatigue detection using multilevel feature extraction and iterative hybrid feature selection. Biomedical Signal Processing and Control 68:102591. https://doi.org/10.1016/j.bspc.2021.102591
Übeyli ED (2010) Lyapunov exponents/probabilistic neural networks for analysis of EEG signals. Expert Systems with Applications 37:985–992. https://doi.org/10.1016/j.eswa.2009.05.078
van den Broek EL, Lisý V, Janssen JH, et al (2010) Affective Man-Machine Interface: Unveiling Human Emotions through Biosignals. In: Fred A, Filipe J, Gamboa H (eds) Biomedical Engineering Systems and Technologies. Springer, Berlin, Heidelberg, pp 21–47
Vapnik VN (1999) An overview of statistical learning theory. IEEE Transactions on Neural Networks 10:988–999. https://doi.org/10.1109/72.788640
Wang M, Huang Z, Li Y, et al (2021) Maximum weight multi-modal information fusion algorithm of electroencephalographs and face images for emotion recognition. Computers & Electrical Engineering 94:107319. https://doi.org/10.1016/j.compeleceng.2021.107319
Wang Q, Wang M, Yang Y, Zhang X (2022) Multi-modal emotion recognition using EEG and speech signals. Computers in Biology and Medicine 149:105907. https://doi.org/10.1016/j.compbiomed.2022.105907
Wang S, Qu J, Zhang Y, Zhang Y (2023) Multimodal Emotion Recognition From EEG Signals and Facial Expressions. IEEE Access 11:33061–33068. https://doi.org/10.1109/ACCESS.2023.3263670
Wang S-H, Li H-T, Chang E-J, Wu A-Y (Andy) (2018) Entropy-Assisted Emotion Recognition of Valence and Arousal Using XGBoost Classifier. In: Iliadis L, Maglogiannis I, Plagianakos V (eds) Artificial Intelligence Applications and Innovations. Springer International Publishing, Cham, pp 249–260
Wang X-W, Nie D, Lu B-L (2011) EEG-Based Emotion Recognition Using Frequency Domain Features and Support Vector Machines. In: Lu B-L, Zhang L, Kwok J (eds) Neural Information Processing. Springer, Berlin, Heidelberg, pp 734–743
Wen T, Zhang Z (2017) Effective and extensible feature extraction method using genetic algorithm-based frequency-domain feature search for epileptic EEG multiclassification. Medicine (Baltimore) 96:e6879. https://doi.org/10.1097/MD.0000000000006879
Wu D, Zhang J, Zhao Q (2020) Multimodal fused emotion recognition about expression-EEG interaction and collaboration using deep learning. IEEE Access 8:133180–133189
Wu X, Zheng W-L, Li Z, Lu B-L (2022) Investigating EEG-based functional connectivity patterns for multimodal emotion recognition. J Neural Eng 19:016012. https://doi.org/10.1088/1741-2552/ac49a7
Xefteris V-R, Tsanousa A, Georgakopoulou N, et al (2022) Graph Theoretical Analysis of EEG Functional Connectivity Patterns and Fusion with Physiological Signals for Emotion Recognition. Sensors 22:8198. https://doi.org/10.3390/s22218198
Xing B, Zhang H, Zhang K, et al (2019) Exploiting EEG Signals and Audiovisual Feature Fusion for Video Emotion Recognition. IEEE Access 7:59844–59861. https://doi.org/10.1109/ACCESS.2019.2914872
Xingyuan W, Chao L, Juan M (2009) Nonlinear dynamic research on EEG signals in HAI experiment. Applied Mathematics and Computation 207:63–74. https://doi.org/10.1016/j.amc.2007.10.064
Yadav SP, Zaidi S, Mishra A, Yadav V (2022) Survey on Machine Learning in Speech Emotion Recognition and Vision Systems Using a Recurrent Neural Network (RNN). Arch Computat Methods Eng 29:1753–1770. https://doi.org/10.1007/s11831-021-09647-x
Yakovyna V, Khavalko V, Sherega V, et al (2021) Biosignal and Image Processing System for Emotion Recognition Applications. In: IT&AS. pp 181–191
Yang K, Wang C, Gu Y, et al (2023) Behavioral and Physiological Signals-Based Deep Multimodal Approach for Mobile Emotion Recognition. IEEE Transactions on Affective Computing 14:1082–1097. https://doi.org/10.1109/TAFFC.2021.3100868
Yang Y, Gao Q, Song Y, et al (2022) Investigating of Deaf Emotion Cognition Pattern By EEG and Facial Expression Combination. IEEE Journal of Biomedical and Health Informatics 26:589–599. https://doi.org/10.1109/JBHI.2021.3092412
Yang Y, Wu Q, Qiu M, et al (2018) Emotion Recognition from Multi-Channel EEG through Parallel Convolutional Recurrent Neural Network. In: 2018 International Joint Conference on Neural Networks (IJCNN). pp 1–7
Yong X, Menon C (2015) EEG classification of different imaginary movements within the same limb. PloS one 10:e0121896
Yu C, Wang M (2022) Survey of emotion recognition methods using EEG information. Cognitive Robotics 2:132–146. https://doi.org/10.1016/j.cogr.2022.06.001

Yu L, Liu H (2003) Feature selection for high-dimensional data: A fast correlation-based filter solution. In: Proceedings of the 20th international conference on machine learning (ICML-03). pp 856–863
Zeng H, Shu X, Wang Y, et al (2021) EmotionCues: Emotion-Oriented Visual Summarization of Classroom Videos. IEEE Transactions on Visualization and Computer Graphics 27:3168–3181. https://doi.org/10.1109/TVCG.2019.2963659
Zhang H (2020) Expression-EEG Based Collaborative Multimodal Emotion Recognition Using Deep AutoEncoder. IEEE Access 8:164130–164143. https://doi.org/10.1109/ACCESS.2020.3021994
Zhang J, Chen M, Zhao S, et al (2016a) ReliefF-Based EEG Sensor Selection Methods for Emotion Recognition. Sensors 16:1558. https://doi.org/10.3390/s16101558
Zhang J, Yin Z, Chen P, Nichele S (2020) Emotion recognition using multi-modal data and machine learning techniques: A tutorial and review. Information Fusion 59:103–126. https://doi.org/10.1016/j.inffus.2020.01.011
Zhang X, Liu J, Shen J, et al (2021a) Emotion Recognition From Multimodal Physiological Signals Using a Regularized Deep Fusion of Kernel Machine. IEEE Transactions on Cybernetics 51:4386–4399. https://doi.org/10.1109/TCYB.2020.2987575
Zhang Y, Cheng C, Zhang Y (2021b) Multimodal Emotion Recognition Using a Hierarchical Fusion Convolutional Neural Network. IEEE Access 9:7943–7951. https://doi.org/10.1109/ACCESS.2021.3049516
Zhang Y, Ji X, Liu B, et al (2017a) Combined feature extraction method for classification of EEG signals. Neural Comput & Applic 28:3153–3161. https://doi.org/10.1007/s00521-016-2230-y
Zhang Y, Ji X, Zhang S (2016b) An approach to EEG-based emotion recognition using combined feature extraction method. Neuroscience letters 633:152–157
Zhang Y, Ji X, Zhang S (2016c) An approach to EEG-based emotion recognition using combined feature extraction method. Neuroscience letters 633:152–157
Zhang Y, Liu B, Ji X, Huang D (2017b) Classification of EEG Signals Based on Autoregressive Model and Wavelet Packet Decomposition. Neural Process Lett 45:365–378. https://doi.org/10.1007/s11063-016-9530-1
Zhang Z (2019) Spectral and time-frequency analysis. EEG Signal Processing and feature extraction 89–116
Zhao L-M, Li R, Zheng W-L, Lu B-L (2019) Classification of Five Emotions from EEG and Eye Movement Signals: Complementary Representation Properties. In: 2019 9th International IEEE/EMBS Conference on Neural Engineering (NER). pp 611–614
Zhao Z-W, Liu W, Lu B-L (2021) Multimodal Emotion Recognition Using a Modified Dense Co-Attention Symmetric Network. In: 2021 10th International IEEE/EMBS Conference on Neural Engineering (NER). pp 73–76
Zheng W-L, Liu W, Lu Y, et al (2019) EmotionMeter: A Multimodal Framework for Recognizing Human Emotions. IEEE Transactions on Cybernetics 49:1110–1122. https://doi.org/10.1109/TCYB.2018.2797176
Zheng X, Yu X, Yin Y, et al (2021) Three-dimensional feature maps and convolutional neural network-based emotion recognition. International Journal of Intelligent Systems 36:6312–6336. https://doi.org/10.1002/int.22551
Zhou J, Wei X, Cheng C, et al (2018) Multimodal Emotion Recognition Method Based on Convolutional Auto-Encoder. International Journal of Computational Intelligence Systems 12:351–358. https://doi.org/10.2991/ijcis.2019.125905651